Banking Networks and the Financial Reallocation Channel of Monetary Policy

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Abstract

This paper studies the role of the banking network in determining the transmission of monetary policy. In an otherwise standard New Keynesian model, credit is supplied by interconnected banks whose market values are linked through cross-holdings. Monetary expansion reallocates credit toward banks whose lending responds most to policy, because their own lending feeds back most strongly into their market value. Whether this reallocation amplifies or dampens monetary transmission depends on whether the responsive banks are also systemically important. A bank is systemically important in this sense when its lending has large spillovers on other banks' ability to lend. The aggregate effect of monetary policy therefore depends on the alignment between which banks respond and which banks matter, a property of the financial network. This alignment can be summarized in a single cross-sectional covariance.

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1 Introduction

Monetary policy changes the quantity of credit in the economy. This paper asks whether it also matters which banks supply that credit. In a standard New Keynesian model, the answer is no: the banking sector is absent or represented by a single intermediary, so a dollar of new lending is a dollar of new lending regardless of who originates it.

Modern banking systems cannot be modeled as a representative intermediary (Bellifemine et al. 2025; Jamilov and Monacelli 2025). Banks are connected through balance sheets, ownership claims, common exposures, and other valuation linkages (Acemoglu et al. 2015; Jackson and Pernoud 2021), and they differ widely in how strongly they respond to a given policy impulse (Drechsler et al. 2017, 2018, 2024). These features, taken together, imply that two equally large credit expansions can have different macroeconomic consequences if they occur at different banks.

We study this question in a New Keynesian model in which credit is supplied by a network of interconnected banks. Each bank holds long-term bonds subject to a value-based leverage constraint, and bank market values are linked through equity cross-holdings in the spirit of Elliott et al. (2014). The household side follows Sims et al. (2023), with patient savers and impatient borrowers, so changes in credit conditions redistribute spending across households. The production side is otherwise standard, and monetary policy operates through both a short-rate rule and central-bank purchases of long-term bonds. Log-linearization reduces the model to a four-equation system in output, inflation, aggregate financial capacity, and the cross-sectional distribution of bank bond holdings. The IS curve and Phillips curve are augmented by aggregate financial capacity, which itself aggregates the cross-section through the network.

Monetary expansion reallocates credit across banks. A rate cut widens the spread between long-bond returns and the deposit rate, making risky long bonds more profitable and pushing every bank to expand its bond holdings. The expansion is uneven, however, because banks' own lending feeds back into their own market values to different degrees. When a bank lends more, the additional profit raises the bank's fundamentals, but cross-holdings allow other banks to claim part of the value gain; only the share that stays with the bank itself relaxes its own leverage constraint. We call this share the bank's *dependency index*. In partial equilibrium, expansionary monetary policy therefore pushes credit toward high-dependency banks, because their lending responds more strongly to a wider credit spread. Whether this reallocation concentrates or disperses credit at the industry level depends on the covariance between bank-level responsiveness and steady-state market share: when the most responsive banks are also the largest, monetary expansion concentrates credit further; when responsive banks tend to be smaller, monetary expansion disperses credit toward them.

Reallocation matters for the aggregate only when a dollar of credit means different

things at different banks. In a banking system without cross-holdings, it would not: every bank's lending would contribute to aggregate capacity in proportion to its market share, and shifting credit across banks would wash out in the aggregate. Cross-holdings break this equivalence. A credit expansion at one bank changes that bank's own market value, but valuation linkages also change the market values of the banks connected to it. When those spillovers are large, a unit of additional lending at this bank does more than enlarge its own balance sheet; it relaxes leverage constraints elsewhere and raises the banking system's overall ability to fund borrowers. We call such a bank systemically important. Systemic importance need not coincide with responsiveness. A high-dependency bank is one whose equity is held mostly by outside investors, so the value gains from its own lending stay with itself and sharpen its response to monetary policy. A systemically important bank, by contrast, is one whose value movements propagate to many other banks through their cross-holdings—which requires that other banks, not outside investors, own a meaningful share of its equity.[1] The same ownership structure therefore tends to pull the two features in opposite directions, and a network in which the most responsive banks also have the largest spillovers is a special configuration rather than a generic property of banking networks.

Combining the two margins gives the paper's central message. Monetary policy raises aggregate financial capacity when responsive banks are also systemically important: when the two coincide, the same monetary impulse delivers a much larger increase in capacity; when they do not, monetary expansion produces a sizable reshuffle of credit across banks but only a modest aggregate response. The aggregate effect of monetary policy therefore depends on the alignment between which banks respond and which banks matter—a property of the financial network, not of any individual bank.

The same alignment shapes the relative output and inflation responses to a given monetary shock. In our log-linearized model, an increase in financial capacity raises borrower spending today relative to expected future spending, which moves output through the demand side. At the same time, shifting consumption from savers to borrowers changes labor supply and marginal cost, which moves inflation. Networks with strong alignment generate larger demand and supply effects from the same monetary impulse; networks with weak alignment generate smaller effects of both.

The mechanism also clarifies why amplification varies across policy tools. Network amplification operates on the part of monetary transmission that runs through bank balance sheets, bond prices, and credit spreads. Conventional rate policy contains such a component, but it also has a direct household Euler-equation channel that bypasses banks. Quantitative easing is more concentrated in the balance-sheet component because

[1] Formally, the dependency index is the diagonal element of the dependency matrix, whereas a bank's systemic importance depends on the entire column of the dependency matrix weighted by other banks' steady-state lending shares.

it works through bond purchases, bond prices, and the distribution of bank portfolios. The same banking network therefore amplifies quantitative easing more strongly than it amplifies the short-rate instrument, without requiring unconventional policy to be more powerful across the board. The result fits the policy experience of the post-2008 and pandemic-era periods, when balance-sheet operations operated alongside conventional rate policy through structurally different transmission channels.

Related literature. The closest macroeconomic benchmark is Sims et al. (2023), who show how a representative intermediary and borrower-saver heterogeneity generate a four-equation New Keynesian system in which credit affects both the IS curve and the Phillips Curve. We inherit this aggregate structure but replace their representative financial intermediary with a financial network of interconnected banks. This makes the cross-section of credit supply a macroeconomic state variable. More broadly, the paper builds on macroeconomic models with financial frictions and heterogeneous agents (Adrian and Shin 2010; Auclert et al. 2025; Brunnermeier and Sannikov 2014; Gertler and Karadi 2011; Gertler and Kiyotaki 2010, 2015; He and Krishnamurthy 2013), including work on monetary reversals (Abadi et al. 2023), central-bank toolkits (Sims and Wu 2021), and conventional versus unconventional policy operations (Bianchi and Bigio 2022; Bigio and Sannikov 2023; Wu and Xie 2025). These models generally aggregate the banking sector; our channel runs through which banks hold marginal credit.

The paper also relates to work on the financial transmission of monetary policy (Bernanke and Gertler 1995; Drechsler et al. 2018), in particular bank heterogeneity and supply-side channels. Bellifemine et al. (2025) show that heterogeneous bank lending propensities amplify aggregate demand, while Jamilov and Monacelli (2025) generates bank wealth heterogeneity from idiosyncratic histories. Our heterogeneity is structural and network-based: balance-sheet linkages determine both the banks that respond to policy and the aggregate importance of their lending. Empirical work documents that monetary policy moves credit spreads (Gertler and Karadi 2015) and that bank-level responses to conventional and unconventional policy are heterogeneous (Jiménez et al. 2014; Kashyap and Stein 2000; Rodnyansky and Darmouni 2017; Wieland and Yang 2020), consistent with the responsiveness margin in our model. Ottonello and Winberry (2020) show that cross-sectional financial positions are state variables for monetary transmission on the borrower side. We obtain an analogous state variable on the intermediary side. On the supply side, Baqaee et al. (2024) derive monetary supply effects from heterogeneous firm pass-through and markups, while Gödl-Hanisch (2025) links bank concentration to Phillips Curve flattening. Our supply effect instead comes from credit-induced consumption redistribution between borrowers and savers, with its strength governed by the banking network.

Finally, the paper contributes to the networks-in-macroeconomics literature. Production-

network models show how input-output linkages make aggregate fluctuations and the relative output and inflation responses depend on network structure (Acemoglu et al. 2012; Carvalho and Tahbaz-Salehi 2019; La'O and Tahbaz-Salehi 2022; Rubbo 2023). Financial-network models study how cross-holdings and interbank linkages transmit distress and amplify shocks (Acemoglu et al. 2015; Elliott et al. 2014). We bring the financial-network object into normal-times monetary transmission rather than crisis propagation. The asymmetric ownership structures we work with are micro-founded by Farboodi (2023), and Glasserman and Young (2016) and Jackson and Pernoud (2021) provide surveys. Banking-network position has been shown empirically to shape credit allocation and shock propagation (Denbee et al. 2021; Diebold and Yılmaz 2014; Kundu and Vats 2024; Li et al. 2023). Adrian and Brunnermeier (2016) further document that a bank's own risk and its systemic contribution are only weakly correlated, consistent with the distinction between bank-level responsiveness and systemic importance that drives our mechanism.

2 A New-Keynesian Model with a Bank Dependency Network

This section develops a New Keynesian model populated by three types of agents: a financial sector of n banks connected through a dependency network (Section 2.1), two households—a patient saver and an impatient borrower (Section 2.2), and a central bank conducting both conventional and unconventional monetary policy (Section 2.3). The production sector follows a standard Calvo pricing structure; its details, together with market clearing conditions and the formal equilibrium definition, are collected in Appendix A.

2.1 The Financial Sector

Consider a discrete-time economy with an infinite horizon, $t = 0, 1, 2, \ldots$ The financial sector consists of n intermediaries (banks) indexed by $i \in \mathcal{N} = \{1, \ldots, n\}$. Banks are risk-neutral and choose portfolios to maximize their market value.

Banks invest in two asset classes. The first is a long-term risky bond modeled as a perpetuity with geometrically decaying coupons: a bond issued at time t pays 1 unit at $t+1$, κ units at $t+2$, κ^2 at $t+3$, and so on, where $\kappa \in [0, 1]$ is the decay parameter. Let Q_t denote the nominal bond price. The gross holding-period return realized at $t+1$ is

$$R^b_{t+1} = \frac{1 + \kappa Q_{t+1}}{Q_t}. \tag{1}$$

The second asset is risk-free reserves $RE_{i,t}$, earning a gross nominal rate R^{re}_t between t and $t+1$. The first order condition for reserve holding implies $\mathbb{E}_t[M^N_{t,t+1}(R^{re}_t - R^s_t)] =$

0, where $M^N_{t,t+1}$ is the nominal stochastic discount factor and R^s_t is the deposit rate. Reserves therefore yield no excess return in expectation, and the bank's problem reduces to managing the risky bond portfolio.

Following Sims et al. (2023) and Wu and Xie (2025), we assume that each banker is born at the beginning of period t, operates for one period, and exits at the beginning of $t+1$ with probability one. Upon exit, the banker distributes terminal wealth as dividends. The newly born FIs receive an exogenous transfer from the unconstrained household including two components: (i) a fixed amount of new equity and (ii) the outstanding long-term bonds held by the previous intermediary. [2]

2.1.1 The Banking Network

The central object in our model is the *dependency matrix* $\mathbf{A}$, which maps bank-level fundamental values to market values and thereby governs how the financial network shapes lending incentives. This subsection derives $\mathbf{A}$ directly in the presence of the structural friction that is central to our analysis.

The network is captured by a matrix $\mathbf{C} \in \mathbb{R}^{n\times n}_+$, where $C_{ij} \in [0,1)$ denotes the fraction of bank j's equity held by bank i. We impose three structural restrictions. First, *no self-ownership*: $C_{ii} = 0$ for all i. Second, *positive outside ownership*: the residual share of bank i held by outside investors (households) is $\hat{C}_{ii} \equiv 1-\sum_{j\in\mathcal{N}} C_{ji} \in (0,1]$ for all i. Third, *convergence*: the spectral radius $\rho(\mathbf{C}) < 1$, guaranteeing that the matrix $(\mathbf{I}-\mathbf{C})^{-1}$ exists and has a convergent Neumann series expansion. Let $\hat{\mathbf{C}} \equiv \mathrm{diag}(\hat{C}_{11},\ldots,\hat{C}_{nn})$ collect the outside ownership shares. The matrix $\hat{\mathbf{C}}$ records the physical fact of who holds equity in whom and does not depend on any friction parameter.

The friction parameter τ. We introduce a structural friction parameter $\tau \in [0,1)$ associated with interbank equity linkages. When dividends flow through cross-holding chains, each link transmits only a fraction $(1-\tau)$ of the dividend it receives. This friction captures two sources of value attenuation in interbank ownership chains. First, *tax leakage*: under U.S. tax law (26 U.S.C. §243), the Dividends Received Deduction for portfolio holdings below 20% is limited to 50%, yielding an effective tax wedge of approximately 10.5% at the current federal corporate rate of 21%. Second, *regulatory capital costs*: Basel III mandates that significant investments in other financial institutions' capital be deducted from CET1 or assigned punitive risk weights, imposing an opportunity cost on cross-holding positions. The conceptual distinction between $\hat{C}_{ii}$ and τ is sharp: the former records the ownership structure—the physical fact of who holds equity in whom—while the latter governs the efficiency of dividend transmission through the network.

[2] While individual bankers turn over each period, the institutional structure of each bank—its position in the financial network and its inherited bond portfolio—persists across generations.

Each bank i maintains a balance sheet consisting of bonds $B_{i,t}$, reserves $RE_{i,t}$, deposits $S_{i,t}$, and an exogenous equity endowment $E_{i,t}$:

$$Q_t B_{i,t} + RE_{i,t} = S_{i,t} + E_{i,t}. \tag{2}$$

The bank's net cash flow (dividend) upon exit at $t+1$ is:

$$D_{i,t+1} = \underbrace{(R^b_{t+1} - R^s_t)\, Q_t B_{i,t}}_{\text{credit spread income}} + \underbrace{(R^{re}_t - R^s_t)\, RE_{i,t}}_{=0 \text{ in expectation}} + \underbrace{R^s_t\, E_{i,t}}_{\text{endowment return}} . \tag{3}$$

The fundamental value of bank i is the expected discounted dividend from its own portfolio operations, absent any network effects:

$$x_{i,t} \equiv \mathbb{E}_t\big[M^N_{t,t+1}\, D_{i,t+1}\big] = \Delta_t\, Q_t B_{i,t} + \bar{x}_{i,t}, \tag{4}$$

where $\Delta_t \equiv \mathbb{E}_t[M^N_{t,t+1}(R^b_{t+1} - R^s_t)]$ is the aggregate credit spread and $\bar{x}_{i,t} \equiv \mathbb{E}_t[M^N_{t,t+1} R^s_t]\, E_{i,t}$ is the value of the endowment. Let $\mathbf{x}_t \equiv (x_{1,t}, \ldots, x_{n,t})'$ collect fundamentals.

A bank's total equity value includes its fundamental assets and the value of its equity stakes in other banks, after accounting for the friction τ that attenuates dividend transmission through each link:

$$\mathbf{V}_t = \mathbf{x}_t + (1-\tau)\, \mathbf{C}\, \mathbf{V}_t, \tag{5}$$

Solving the fixed-point equation:

$$\mathbf{V}_t = \big[\mathbf{I} - (1-\tau)\mathbf{C}\big]^{-1} \mathbf{x}_t = \left(\sum_{k=0}^{\infty} (1-\tau)^k \mathbf{C}^k \right) \mathbf{x}_t. \tag{6}$$

The Neumann series has a transparent economic interpretation: $(1-\tau)^k \mathbf{C}^k \mathbf{x}_t$ captures the value flowing to each bank through ownership chains of exactly k links, attenuated by a factor $(1-\tau)^k$. A direct holding ($k=1$) transmits value with one round of friction; an indirect holding through an intermediary ($k=2$) transmits value with two rounds of friction; and so on.

Market value and the dependency matrix. Outside investors hold a fraction $\hat{C}_{ii}$ of each bank's total equity. The market value of bank i's outside equity—the value relevant for the bank's objective and its leverage constraint—is:

$$\mathbf{v}_t = \hat{\mathbf{C}}\, \mathbf{V}_t = \underbrace{\hat{\mathbf{C}}\, \big[\mathbf{I} - (1-\tau)\mathbf{C}\big]^{-1}}_{\equiv \mathbf{A}(\tau)} \mathbf{x}_t. \tag{7}$$

Definition 1 (Dependency Matrix)**.** *The* dependency matrix $\mathbf{A}(\tau) \equiv \hat{\mathbf{C}}\,[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ *maps fundamental values to market values. Its element A_{ij} measures the marginal effect of a unit increase in bank j's fundamental value on bank i's market value. The diagonal element A_{ii}, which we term bank i's* dependency index*, captures how much of the bank's own fundamental value is retained in its market value after accounting for all network interactions and frictions.*

Intuitively, A_{ii} tells us how much of one extra dollar of bank i's own profit it ultimately keeps as market value. Two opposing forces shape this number. The first is *dilution*: if other banks own a large share of bank i, a sizeable fraction of i's value flows out through the ownership chain and is captured by those outside holders rather than by i itself. The second is *network feedback*: some of the value that flows out eventually returns to i through closed ownership loops, because i in turn holds stakes in banks that hold stakes in i. The friction τ governs how much of that returning value survives the round trip. The dependency index is the net of these two forces, and the balance differs systematically across banks: core banks heavily held by others have a small dilution factor but, when τ is small, may recover much of the loss through dense feedback loops; peripheral banks lose little to dilution to begin with but also benefit from little feedback. For notational economy, we suppress the dependence on τ except where comparative statics are explicitly discussed.

The Neumann expansion makes the two-force decomposition transparent:

$$A_{ii} = \underbrace{\hat{C}_{ii}}_{\text{dilution factor}} \times \underbrace{\left[1 + (1-\tau)^2 (C^2)_{ii} + (1-\tau)^3 (C^3)_{ii} + \cdots\right]}_{\text{amplification factor}}, \tag{8}$$

where the $k = 1$ term vanishes because $C_{ii} = 0$ and $(C^k)_{ii} = \sum_{j_1,\ldots,j_{k-1}} C_{ij_1} C_{j_1 j_2} \cdots C_{j_{k-1} i}$ sums over all ownership paths of length k beginning and ending at bank i. In the frictionless benchmark ($\tau = 0$) the amplification factor operates at full force and can push A_{ii} above one for banks at the center of dense loops. As τ rises, each round of feedback is attenuated by $(1-\tau)^k$; in the limit $\tau \to 1$, $\mathbf{A} \to \hat{\mathbf{C}}$ and the dependency index collapses to the outside ownership share. The empirically relevant regime is intermediate: τ is large enough to keep the feedback loop convergent but small enough that the dispersion of A_{ii} across banks remains economically meaningful.

In the log-linear equilibrium system, the level derivative $\mathbf{A}$ is converted into the valuation-elasticity matrix

$$\mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}, \qquad \hat{v}_{i,t} = \sum_j \mathcal{A}_{ij} \hat{x}_{j,t}, \tag{9}$$

so that $\mathbf{A}$ is the level dependency matrix while $\mathcal{A}$ is the object entering the log-linear four-equation system.

The friction τ destroys aggregate value, with the dissipation concentrated at the most interconnected institutions. The following result quantifies this dissipation.

Proposition 1 (Value Dissipation)**.** *For any cross-holdings matrix* $\mathbf{C}$ *satisfying the structural restrictions above and any* $\tau \in [0, 1)$*:*

$$\sum_{i \in \mathcal{N}} A_{ij}(\tau) = 1 - \tau \sum_{i \in \mathcal{N}} \left[\mathbf{C}\,(\mathbf{I} - (1-\tau)\mathbf{C})^{-1}\right]_{ij} \leq 1 \quad \textit{for all } j \in \mathcal{N}, \tag{10}$$

with equality if and only if $\tau = 0$ *or* $\mathbf{C} = \mathbf{0}$*.*

Proof. See Appendix B.1. □

One dollar of bank j's fundamental value generates less than one dollar of aggregate market value across the banking system whenever $\tau > 0$; the shortfall equals the cumulative tax and regulatory leakage incurred as dividends flow through all direct and indirect ownership chains originating from bank j. In the frictionless case ($\tau = 0$) the column sums equal one exactly, and the network only redistributes value across banks without dissipating it.

When $\tau > 0$, the dependency indices A_{ii} are heterogeneous across banks. This heterogeneity is what allows monetary policy to reallocate credit across banks (Proposition 3) and what makes that reallocation matter for aggregate marginal costs in general equilibrium.

2.1.2 Individual Bank Problem

Each bank i chooses bond holdings $B_{i,t}$ to maximize its market value $v_{i,t}$, subject to a leverage constraint that ties borrowing capacity to the bank's network-determined market value and penalizes portfolio adjustment:

$$Q_t B_{i,t} \leq \phi_{i,t}\, v_{i,t} - \eta_B\, \bar{Q}\, (B_{i,t} - B_{i,t-1}), \tag{11}$$

where $\phi_{i,t} > 0$ is an idiosyncratic credit-capacity shock and $\eta_B \geq 0$ is a portfolio adjustment cost parameter. The adjustment cost term penalizes changes in bond holdings relative to the previous period, reflecting regulatory reporting costs, market impact in illiquid bond markets, and internal risk-management frictions (Gertler and Kiyotaki 2010). The cost is scaled by the steady-state bond price $\bar{Q}$ for dimensional consistency and vanishes in steady state ($B_{i,t} = B_{i,t-1}$), so it does not affect the long-run equilibrium. This specification creates a value feedback loop: lending decisions affect fundamental values through $x_{i,t} = \Delta_t\, Q_t B_{i,t} + \bar{x}_{i,t}$, which propagate through the network to affect market values $v_{i,t} = \sum_j A_{ij}\, x_{j,t}$, which in turn determine the tightness of the constraint. The strength of this internal feedback depends on the bank's dependency index A_{ii}.

To obtain a closed-form solution, we impose a strategic restriction that is the financial-network analogue of taking competitors' output as given in a Cournot oligopoly. Each bank recognizes that its own lending feeds back to its own valuation through the network, but does not attempt to manipulate other banks' valuations strategically. This is exact when banks are small relative to the network and becomes a close approximation when the number of banks is large.

Assumption 1 (Fixed Spillovers)**.** *Each bank i internalizes the effect of its lending on its own valuation through the self-loop A_{ii}, but treats the value contributions from other banks as exogenous to its portfolio choice. Formally, bank i decomposes its market value as:*

$$v_{i,t} = A_{ii}\big(\Delta_t\, Q_t B_{i,t} + \bar{x}_{i,t}\big) + \bar{v}_{i,t}, \tag{12}$$

where $\bar{v}_{i,t} \equiv \sum_{j\neq i} A_{ij}\, x_{j,t}$ is taken as given.

Under Assumption 1, bank i solves:

$$\max_{B_{i,t}\geq 0}\; v_{i,t} = A_{ii}\big(\Delta_t\, Q_t B_{i,t} + \bar{x}_{i,t}\big) + \bar{v}_{i,t} \quad \text{s.t.} \quad Q_t B_{i,t} \leq \phi_{i,t}\, v_{i,t} - \eta_B\, \bar{Q}\,(B_{i,t} - B_{i,t-1}). \tag{13}$$

Since the objective is linear and increasing in $B_{i,t}$ (given $\Delta_t > 0$), risk-neutral banks lever up until the constraint binds. Substituting the binding constraint and solving yields the optimal credit supply:

$$B^*_{i,t} = \frac{\phi_{i,t}\big(A_{ii}\,\bar{x}_{i,t} + \bar{v}_{i,t}\big) + \eta_B\,\bar{Q}\,B_{i,t-1}}{Q_t\big(1 - \phi_{i,t}\,A_{ii}\,\Delta_t\big) + \eta_B\,\bar{Q}}. \tag{14}$$

The adjustment cost parameter η_B introduces inertia: current lending depends positively on the inherited portfolio $B_{i,t-1}$. In steady state, $B_{i,t} = B_{i,t-1}$ and the adjustment cost term cancels from both sides, so the steady-state allocation is identical to the frictionless case ($\eta_B = 0$). For the analytical results that follow, we present expressions in the $\eta_B = 0$ case for clarity; all results extend to $\eta_B > 0$.[3]

Throughout, we restrict attention to parameter regions where the value feedback loop converges rather than explodes—the bank-level analogue of requiring that a geometric series have ratio less than one. Concretely, $1 - \phi_{i,t}\,A_{ii}\,\Delta_t > 0$ for all i and t, and in the log-linear system the matrix feedback is bounded by $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$.

2.1.3 Credit Supply in Partial Equilibrium

Before embedding the banking sector into general equilibrium, we isolate the micro-level mechanisms through which the network structure shapes credit supply. Throughout this

[3] With $\eta_B > 0$, the feedback multiplier $(1 - \phi_{i,t}A_{ii}\Delta_t)^{-1}$ is replaced by $(1 - \phi_{i,t}A_{ii}\Delta_t + \eta_B\bar{Q}/Q_t)^{-1}$, which dampens contemporaneous amplification but preserves the comparative static $\partial\mathcal{Z}_{i,t}/\partial A_{ii} > 0$.

subsubsection, we take asset prices $\{Q_t\}$, the nominal stochastic discount factor $M^N_{t,t+1}$, and the aggregate credit spread Δ_t as exogenous, and derive analytical results that characterize how the dependency matrix shapes individual bank responses and aggregate credit market structure. These partial-equilibrium insights—combined with the household and monetary policy blocks that follow—determine the full general equilibrium dynamics analyzed in Sections 3–4.

Definition 2 (Partial Equilibrium)**.** *Given the network structure* $\mathbf{A}$*, exogenous processes* $\{\Delta_t, \phi_{i,t}, \bar{x}_{i,t}\}$*, and asset prices* $\{Q_t\}$*, a partial equilibrium is a collection of portfolio choices* $\{B^*_{i,t}\}_{i\in\mathcal{N}}$ *and valuations* $\{\mathbf{v}_t, \mathbf{x}_t\}$ *such that: (i) each bank* i *chooses* $B^*_{i,t}$ *according to* (14)*; (ii) fundamental values satisfy* $x_{i,t} = \Delta_t\, Q_t B^*_{i,t} + \bar{x}_{i,t}$*; and (iii) market values satisfy* $\mathbf{v}_t = \mathbf{A}\,\mathbf{x}_t$*.*

The closed-form solution (14) reveals that the network enters each bank's credit decision through a single channel: the dependency index A_{ii} appears in the denominator as part of a feedback multiplier $\big(1 - \phi_{i,t}\, A_{ii}\, \Delta_t\big)^{-1}$. Banks with higher A_{ii} face a more powerful feedback loop—each dollar of lending generates more market value, which relaxes the leverage constraint by more, enabling more lending—and therefore respond more aggressively to changes in the credit spread.

Proposition 2 (Value Feedback Channel)**.** *Under Assumption 1 and the stability condition in Section 2.1.2, the semi-elasticity of bank* i*'s credit supply with respect to the aggregate credit spread* Δ_t *is strictly positive and strictly increasing in the dependency index* A_{ii}*. Banks that retain a larger share of their own fundamental value respond more aggressively to the same policy impulse.*

Proof. See Appendix B.2. The formal expression is $\mathcal{Z}_{i,t} = \phi_{i,t}A_{ii}/(1 - \phi_{i,t}A_{ii}\Delta_t)$, which is manifestly positive and increasing in A_{ii}. □

The economic mechanism operates as follows. An expansionary monetary policy that widens the credit spread Δ_t increases the expected profitability of lending for all banks. For a standalone bank ($A_{ii} = 1$, no network), this raises its fundamental value, which relaxes its leverage constraint by a proportional amount, enabling some additional lending. For a bank embedded in the network, the initial gain in fundamental value propagates through the dependency network to partner banks, whose higher valuations in turn raise the value of bank i's equity stakes in them, feeding the gain back to bank i. The strength of this self-reinforcing loop depends on A_{ii}: it captures how efficiently the feedback circuit converts a bank's own lending into its own market value, after accounting for all the value that leaks outward to other holders along the way.

In general equilibrium, Proposition 2 provides the micro-foundation for two aggregate phenomena. First, the monotonicity $\partial\mathcal{Z}_{i,t}/\partial A_{ii} > 0$ helps generate credit reallocation across banks that carry different log-effective weights $\tilde{\omega}^{\ell}_j$. Second, inherited credit positions create persistent financial wedges in the aggregate system.

Policy-induced credit reallocation. We now examine how the heterogeneous bank responses identified in Proposition 2 reshape the aggregate market structure. Let aggregate credit supply be $B_t \equiv \sum_{i\in\mathcal{N}} B_{i,t}$, and define bank i's market share as $\theta_{i,t} \equiv B_{i,t}/B_t$. We measure concentration using the Herfindahl–Hirschman Index: $HHI_t \equiv \sum_{i\in\mathcal{N}} \theta_{i,t}^2$.

For any cross-sectional variables $\{y_{i,t}\}$ and $\{z_{i,t}\}$, write the share-weighted expectation and covariance as

$$\mathbb{E}_{\theta,t}[y_{i,t}] \equiv \sum_{i\in\mathcal{N}} \theta_{i,t}\, y_{i,t}, \tag{15}$$

$$\mathrm{Cov}_{\theta,t}(y_{i,t}, z_{i,t}) \equiv \sum_{i\in\mathcal{N}} \theta_{i,t}\big(y_{i,t} - \mathbb{E}_{\theta,t}[y_{i,t}]\big)\big(z_{i,t} - \mathbb{E}_{\theta,t}[z_{i,t}]\big). \tag{16}$$

Proposition 3 (Policy-Induced Credit Reallocation)**.** *Accommodative monetary policy ($d\Delta_t > 0$) reshapes the cross-sectional distribution of credit. Market share flows toward banks whose dependency index exceeds the share-weighted average, and away from banks whose index falls below it. The net effect on market concentration is:*

$$dHHI_t = 2 \cdot \mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t}) \cdot d\Delta_t, \tag{17}$$

where $\mathcal{Z}_{i,t}$ is the semi-elasticity from Proposition 2. Since $\mathcal{Z}_{i,t}$ is strictly increasing in A_{ii}, concentration rises if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$—that is, if and only if larger banks tend to have higher dependency indices.

Proof. See Appendix B.3. □

Proposition 3 establishes that monetary policy is not distributionally neutral: it actively reshapes the industrial organization of the banking sector. An expansionary shock stimulates all banks, but those with higher dependency indices expand disproportionately. Whether this concentrates or disperses credit depends on the sign of $\mathrm{Cov}_\theta(\theta_i, A_{ii})$, which the regime analysis above pins down: in the frictionless benchmark core banks have the highest A_{ii}, while in the empirically relevant high-τ regime the dilution factor dominates and the covariance can flip sign.

Corollary 4 (Market Share Reallocation)**.** *In response to a marginal increase in the credit spread $d\Delta_t > 0$, bank i's market share grows ($d\theta_{i,t} > 0$) if and only if its semi-elasticity exceeds the share-weighted average:*

$$d\ln\theta_{i,t} = \big(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\big)\, d\Delta_t, \tag{18}$$

where $\bar{\mathcal{Z}}_t \equiv \mathbb{E}_{\theta,t}[\mathcal{Z}_{j,t}]$. Since $\mathcal{Z}_{i,t}$ is increasing in A_{ii}, capital flows toward banks with higher dependency indices.

The key implication for general equilibrium is that the cross-sectional credit distribution is endogenous to monetary policy, making the vector $\hat{\mathbf{b}}_t^{FI}$ an independent macroeconomic state variable. Aggregated through log-effective network weights $\tilde{\omega}_j^{\ell}$, this distribution determines the output-inflation trade-off: the financial architecture of the banking sector shapes the effectiveness of monetary policy through a channel entirely absent from representative-intermediary models.

Network topology and the amplification-reallocation decomposition. The preceding results show that the network affects monetary transmission through two distinct channels: *amplification* (all banks expand more due to value feedback) and *reallocation* (credit shifts across banks with different dependency indices). A natural question is whether these two effects always move together, or whether network architectures exist that activate one without the other. The answer depends on the symmetry of the network.

Proposition 5 (Symmetric vs. Asymmetric Networks)**.** *Consider the class of networks parameterized by a uniform cross-holding intensity $\delta \in [0, 1)$ and a topology matrix $\mathbf{G}$ with $G_{ii} = 0$ and $\sum_j G_{ij} = 1$ for all i, so that $\mathbf{C} = \delta\mathbf{G}$.*

1. Symmetric networks. *If $\mathbf{G}$ is vertex-transitive (each bank occupies the same network position), then $A_{ii} = A_{jj}$ for all i, j. The network amplifies aggregate credit supply but does not generate reallocation: $dHHI_t = 0$ for a common $\phi_{i,t} = \bar{\phi}$.*

2. Asymmetric networks. *If banks occupy non-equivalent network positions, then dependency indices differ generically. The network activates both the amplification and reallocation channels, with the direction of reallocation governed by $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii})$.*

Proof. See Appendix B.4. □

Proposition 5 provides the theoretical anchor for distinguishing the effects of network *density* (the scalar δ) from network *asymmetry* (the symmetry properties of $\mathbf{G}$). Both dense vertex-transitive networks (e.g., a complete graph) and dense asymmetric networks (e.g., a core-periphery structure) amplify aggregate credit, since higher δ raises the average feedback multiplier. But only asymmetric networks generate differential dependency indices across banks and thereby activate the reallocation mechanism.

Among standard topologies with $\delta = 0.15$ and $n = 5$, complete and ring graphs are vertex-transitive (case a, A_{ii} uniform across banks), while star and core-periphery configurations are not (case b, A_{ii} differs between center/core and periphery). Only case (b) generates the reallocation channel.

This decomposition carries a sharp implication for general equilibrium: symmetric networks can only amplify the demand channel of monetary transmission, while the

supply-side reallocation channel—operating through effective market share wedges and Phillips Curve flattening—requires network asymmetry.

A retrospective examination of the four results above reveals a striking feature of their proofs: none invokes the specific functional form $\mathbf{A} = \hat{\mathbf{C}}(\mathbf{I} - (1-\tau)\mathbf{C})^{-1}$. Every proposition relies only on the following general properties of $\mathbf{A}$:

1. $A_{ij} \geq 0$ for all i, j (non-negative value transmission);
2. $A_{ii} \in (0, 1]$ with cross-sectional heterogeneity (differential self-retention);
3. column sums $\sum_i A_{ij} \leq 1$, with strict inequality when $\tau > 0$ (value dissipation);
4. the Neumann expansion $\sum_k [(1-\tau)\mathbf{C}]^k$ converges (stability).

The dependency matrix $\mathbf{A}$ is therefore a *sufficient statistic* for the banking network's effect on credit supply: any micro-foundation that generates a matrix with these four properties will produce identical analytical results. Equity cross-holdings are one such micro-foundation—chosen for their parsimony and the precedent of Elliott et al. (2014)—but they are not the only one. In practice, bank interdependence arises from multiple channels: interbank lending, common asset exposures (Allen et al. 2012), derivatives counterparty relationships, deposit market spillovers, information contagion, and fire-sale externalities. The key empirical question is not whether any single channel dominates, but whether the *aggregate* dependency matrix—the net effect of all channels—exhibits the four properties listed above.

A large empirical literature has estimated objects that are the reduced-form counterparts of our structural $\mathbf{A}$, confirming that it possesses all the required properties. Adrian and Brunnermeier (2016) measure systemic risk contributions through CoVaR and find that individual banks' VaR and ΔCoVaR are only weakly correlated in the cross section—direct evidence that effective network weights $\tilde{\omega}_j^\ell$ differ from market shares ω_j. Diebold and Yılmaz (2014) construct connectedness measures from variance decompositions and document that system-wide connectedness is quantitatively large (approximately 60% in normal times, exceeding 90% during crises), with a pronounced core-periphery asymmetry. Billio et al. (2012) use Granger-causality networks to show that financial institutions have become increasingly interconnected, with banks and insurance companies at the center of the network. These estimates consistently confirm that the aggregate bank dependency structure is quantitatively significant, asymmetric with a core-periphery pattern, and time-varying with the monetary policy stance—precisely the properties on which our theory relies.

Closing the model with households and a monetary authority lets us answer two questions left open by partial equilibrium: how the cross-sectional reallocation feeds back to aggregate prices, and whether the network shifts the composition rather than only the level of monetary transmission.

2.2 Households

Following Sims et al. (2023), the household sector consists of two types of agents: a representative unconstrained household (patient) and a representative constrained household (impatient). The patient household is the standard optimizing agent in New Keynesian models: it consumes, supplies labor, saves through short-term deposits, and owns equity claims on both production firms and financial intermediaries. The impatient household is less patient, does not supply labor, and finances its consumption by issuing long-term bonds. Let $z_N \in [0, 1)$ denote the population share of impatient households. The log-linear aggregate equations use the borrower steady-state consumption share $z_C \equiv \bar{C}_B/\bar{Y}$, which need not equal z_N away from a normalization with equal per-capita steady-state consumption.

2.2.1 The Patient Household

The representative patient household has preferences over consumption $C_{P,t}$ and labor L_t. It discounts future utility flows by the subjective discount factor $\beta \in (0, 1)$. The patient household's lifetime utility is given by:

$$\mathcal{V}_t^P = \max\ \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{P,t+j}^{1-\sigma} - 1}{1-\sigma} - \psi \frac{L_{t+j}^{1+\chi}}{1+\chi} \right], \tag{19}$$

where $\sigma > 0$ is the inverse of the intertemporal elasticity of substitution, $\chi \geq 0$ is the inverse Frisch elasticity of labor supply, and $\psi > 0$ is a scaling parameter governing the disutility of labor.

The patient household has access to several financial instruments and ownership claims. It saves through one-period nominal deposits $S_{i,t}$ held with financial intermediary i. These deposits pay a gross nominal interest rate R_t^s between periods t and $t+1$. The patient household owns all equity in the production sector and receives aggregate dividends $\bar{D}_t^F$ from final goods producers, retailers, and wholesale firms. It also holds the outside equity claims on the banking sector: recall from Section 2.1.1 that the underlying ownership structure is captured by $\hat{C}_{ii} \in (0, 1]$, representing the fraction of bank i's equity held by households. The patient household receives dividends $P_t D_t^{P,FI}$ from this outside equity position. Besides, it makes a transfer X_t^b to the impatient household each period, as well as a transfer $P_t X_{i,t}^{FI}$ to each financial intermediary. Lastly, the patient household receives a lump-sum transfer T_t from the fiscal authority, which includes the central bank's operating surplus remitted to the treasury.

Treating all prices, dividends from firms, government transfers, the transfer schedule to the impatient household, and the equity injection schedule to banks as given, the representative patient household maximizes Equation (19) by choosing a sequence of consumption $C_{P,t}$, labor supply L_t, and one-period deposits $\{S_{i,t}\}_{i \in \mathcal{N}}$, subject to the

budget constraint

$$P_tC_{P,t}+\sum_{i\in\mathcal{N}}S_{i,t}+\sum_{i\in\mathcal{N}}P_tX_{i,t}^{FI}+P_tX_t^b\leq W_tL_t+\sum_{i\in\mathcal{N}}R_{t-1}^sS_{i,t-1}+P_tD_t^F+P_tD_t^{P,FI}+P_tT_t. \quad (20)$$

The patient household's nominal stochastic discount factor (SDF), which prices nominal payoffs from the perspective of the household sector, is defined as:

$$M_{t,t+1}^N\equiv\beta\left(\frac{C_{P,t+1}}{C_{P,t}}\right)^{-\sigma}\frac{1}{\Pi_{t+1}}. \quad (21)$$

This SDF determines nominal asset prices, including the valuation of banks' bond portfolios and the equilibrium credit spread that were taken as exogenous in Section 2.1.3.

2.2.2 The Impatient Household

The representative impatient household derives utility solely from consumption $C_{B,t}$ and does not supply labor. The impatient household is less patient than the patient household, discounting future utility flows by $\beta_B<\beta$. This relative impatience generates a borrowing motive that sustains the demand for long-term credit in equilibrium. The impatient household's lifetime utility is given by:

$$\mathcal{V}_t^B=\max\ \mathbb{E}_t\sum_{j=0}^{\infty}\beta_B^j\left[\frac{(C_{B,t+j})^{1-\sigma}-1}{1-\sigma}\right]. \quad (22)$$

We assume the same intertemporal elasticity of substitution $1/\sigma$ for both household types, which simplifies aggregation while preserving the essential heterogeneity in patience that drives credit demand.

Each period, the impatient household must finance consumption expenditure $P_tC_{B,t}$ plus coupon payments on outstanding debt B_{t-1}^C using proceeds from new bond issuance and the transfer received from the patient household. The impatient household's nominal budget constraint is:

$$P_tC_{B,t}+B_{t-1}^C\leq Q_t\left(B_t^C-\kappa B_{t-1}^C\right)+P_tX_t^b, \quad (23)$$

where the left-hand side represents total uses of funds (consumption plus debt service) and the right-hand side represents total sources (new borrowing plus the transfer from the patient household).

We assume that the transfer from the patient to the impatient household is set according to

$$P_tX_t^b=(1+\kappa Q_t)B_{t-1}^C, \quad (24)$$

which implies that the patient household effectively pays off the impatient household's

maturing debt obligations each period. Substituting into the budget constraint (23) and simplifying, we obtain that borrower consumption is determined by a static relationship:

$$P_t C_{B,t} = Q_t B_t^C. \tag{25}$$

The borrower's real consumption therefore equals the real market value of its outstanding debt.

This transfer rule serves as a simplifying device that eliminates B_{t-1}^C as a relevant state variable for the impatient household's optimization, reducing the dimensionality of the system. It ensures that borrower consumption dynamics are driven entirely by current credit conditions rather than by the history of debt accumulation, allowing us to isolate the effects of financial frictions from intertemporal debt dynamics.

Although the constraint appears static, the impatient household's Euler equation involves expectations of future bond returns because current bond issuance B_t^C determines next period's coupon obligations, which affect the terms at which debt can be rolled over. To ensure stationarity of the debt position, we introduce a small portfolio holding cost following Schmitt-Grohé and Uribe (2003): the impatient household's optimality condition takes the form

$$1 = \beta_B \left(\frac{C_{B,t+1}}{C_{B,t}} \right)^{-\sigma} \frac{R_{b,t+1}}{\Pi_{t+1}} - \psi_b \left(\frac{b_{C,t}}{\bar{b}_C} - 1 \right),$$

where $\psi_b \geq 0$ is a small debt-elasticity parameter. This term penalizes deviations of borrower debt from its steady-state level, preventing the near-unit-root dynamics that arise in incomplete-markets models. The term vanishes in steady state ($b_{C,t} = \bar{b}_C$) and is kept in the bond-pricing closure rather than appended directly to the benchmark IS curve in Section 3.1.

The impatient household plays a crucial role in generating a meaningful demand for long-term credit intermediated by the banking sector. In the absence of such demand, financial intermediaries would have no profitable lending opportunities, and the dependency network would be irrelevant for equilibrium dynamics. The impatient household's relative impatience ($\beta_B < \beta$) ensures that in steady state it is a net borrower while the patient household is a net saver, with financial intermediaries channeling funds between the two household types. From an economic interpretation perspective, the impatient household can be viewed as a reduced-form representation of the investment sector or the market for durable consumption goods; the dynamics of borrower consumption in this framework align closely with the behavior of investment in richer models with physical capital accumulation (Sims and Wu 2021).

2.3 Central Bank and Monetary Policy

The central bank conducts monetary policy through two instruments: a short-term policy rate and a portfolio of long-term bonds. This dual-instrument framework allows us to study both conventional interest rate policy and unconventional balance sheet policies (quantitative easing).

The central bank holds a portfolio of long-term bonds B_t^{cb} issued by the impatient household, financed by the creation of reserves RE_t. The central bank's balance sheet identity is:

$$Q_t B_t^{cb} = RE_t. \tag{26}$$

We define the real value of the central bank's bond portfolio as $QE_t \equiv Q_t B_t^{cb}/P_t = q_t b_t^{cb}$, where $q_t \equiv Q_t/P_t$ is the real bond price and $b_t^{cb} \equiv B_t^{cb}/P_t$ is real bond holdings. The central bank can freely adjust its bond portfolio, which constitutes the quantitative easing (QE) instrument.

The short-term nominal interest rate R_t^s follows a Taylor-type rule that responds to deviations of inflation and output from their targets:

$$\frac{R_t^s}{R^s} = \left(\frac{R_{t-1}^s}{R^s}\right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{\phi_y}\right]^{1-\rho_r} e^{\varepsilon_{r,t}}, \tag{27}$$

where R^s is the steady-state nominal interest rate, Π^* is the inflation target, Y_t^* is potential output, $\rho_r \in [0,1)$ captures interest rate smoothing, $\phi_\pi > 1$ and $\phi_y \geq 0$ are the response coefficients to inflation and the output gap, and $\varepsilon_{r,t}$ is a monetary policy shock.

The central bank's bond portfolio follows a similar rule:

$$\frac{QE_t}{QE} = \left(\frac{QE_{t-1}}{QE}\right)^{\rho_q} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{-\lambda_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{-\lambda_y}\right]^{1-\rho_q} e^{\varepsilon_{q,t}}, \tag{28}$$

where QE is the steady-state bond portfolio, $\rho_q \in [0,1)$ is the smoothing parameter, $\lambda_\pi \geq 0$ and $\lambda_y \geq 0$ govern the response to inflation and output, and $\varepsilon_{q,t}$ is a QE shock. The negative signs reflect that the central bank expands its bond portfolio (easing) when inflation or output fall below target.

Finally, the central bank pays interest R_t^{re} on reserves held by financial intermediaries. We assume that the central bank sets the interest on reserves equal to the short-term policy rate:

$$R_t^{re} = R_t^s. \tag{29}$$

This assumption, combined with the first order condition of reserve holdings from the financial intermediaries' problem, implies that banks are indifferent between holding reserves and short-term deposits, and that reserves yield no excess return in expectation.

The central bank earns returns on its bond portfolio and pays interest on reserves. The operating surplus (or deficit) is remitted to the fiscal authority, which then transfers it to the patient household. This transfer satisfies:

$$P_t T_t = R_t^b Q_{t-1} B_{t-1}^{cb} - R_{t-1}^{re} RE_{t-1}, \tag{30}$$

where $R_t^b Q_{t-1} B_{t-1}^{cb}$ is the return on the central bank's bond portfolio (coupon plus capital gain) and $R_{t-1}^{re} RE_{t-1}$ is the interest paid on reserves. When the surplus is negative, the transfer becomes a lump-sum tax on the patient household. This specification ensures that the government budget constraint is satisfied and that central bank operations do not create fiscal imbalances.

3 Aggregation and the Role of Effective Network Weights

Although the banking block of Section 2 contains n heterogeneous banks, the demand side of the economy interacts with this block through a single scalar—what we will call the *net financial capacity* of the banking sector. The goal of this section is twofold: first, to derive the four-equation system that summarizes general equilibrium dynamics; second, and more importantly, to make precise the sense in which a dollar of new lending at different banks contributes unequally to aggregate financial capacity.

We now log-linearize the general equilibrium model and reduce it to a compact set of four structural equations. Throughout, hatted variables $\hat{x}_t \equiv \ln(x_t/\bar{x})$ denote log-deviations from steady state. The level dependency matrix $\mathbf{A}$ defined in Section 2.1.1 is not itself a log-linear elasticity; the log-linearization derivations, including the conversion to the elasticity matrix $\mathcal{A}$ used throughout this section, are collected in Appendix A.

3.1 Financial Capacity and Aggregate Demand

What matters for borrower spending today is how much of borrower debt the banking system is willing to absorb at current asset prices. We summarize this with the *net financial capacity* of the banking sector,

$$\Theta_t^{\text{net}} \equiv \hat{q}_t + \hat{b}_t^{FI}, \tag{31}$$

which combines the real bond price and aggregate private lending into a single scalar. The analogous central-bank object is the real QE variable $\widehat{qe}_t \equiv \hat{q}_t + \hat{b}_t^{cb}$. The two capacity measures share $\hat{q}_t$—a higher bond price raises the market value of bonds in either holder's portfolio—but keep the holders separate so that QE and bank lending can be tracked independently in the IS curve below.

Log-linearizing goods market clearing, borrower consumption, and the patient household's Euler equation yields:

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{1-z_C}{\sigma}\big(\hat{r}^s_t - \mathbb{E}_t\hat{\pi}_{t+1}\big) + z_C\bar{b}^{FI}\big(\Theta^{\text{net}}_t - \mathbb{E}_t\Theta^{\text{net}}_{t+1}\big) + z_C\bar{b}^{CB}\big(\widehat{qe}_t - \mathbb{E}_t\widehat{qe}_{t+1}\big), \quad (32)$$

where $z_C \equiv \bar{C}_B/\bar{Y}$ is the borrower's steady-state consumption share, and $\bar{b}^{FI}$ and $\bar{b}^{CB}$ are the steady-state shares of borrower debt held by intermediaries and the central bank.[4]

The IS curve has three economically distinct components. The first is the standard intertemporal substitution term: a higher real rate today shifts saver consumption from the present to the future. The remaining two are the financial channels. Both load on *changes* in capacity rather than levels, because borrower spending responds to current credit conditions relative to expected future conditions: a purely permanent expansion of capacity has no intertemporal bite, while a temporary expansion shifts borrower spending forward. The two financial terms also make explicit that QE and conventional lending enter symmetrically as sources of borrower debt absorption, weighted by the steady-state share each holder absorbs.

3.2 Financial Capacity and Marginal Cost

With Calvo pricing frictions, the augmented Phillips Curve is:

$$\hat{\pi}_t = \beta\,\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\,\hat{y}_t - \Lambda_C\big(\bar{b}^{FI}\,\Theta^{\text{net}}_t + \bar{b}^{CB}\,\widehat{qe}_t\big) - \psi_a\,\hat{a}_t. \quad (33)$$

Two terms differ from the textbook NKPC. First, financial capacity enters with a *negative* coefficient: when capacity rises, borrowers consume more for a given output level, so savers consume less and supply more labor at the margin. The lower saver marginal utility of consumption translates into a lower real wage demand and hence a lower marginal cost. This is the supply-side counterpart of the financial channel in the IS curve, and it is what allows credit expansion to generate output without proportional inflation pressure. Second, the slope κ_y on the output gap is steeper than in the representative-agent benchmark because consumption is concentrated on savers ($1 - z_C < 1$), making their marginal utility more sensitive to aggregate output. Unlike the IS curve, the NKPC loads on the *level* of capacity, not its change: marginal cost responds to the current borrower–saver consumption split, regardless of whether the shift is permanent or transitory.

The composite coefficients are

$$\kappa_y \equiv \lambda_p\left(\chi + \frac{\sigma}{1-z_C}\right), \quad \Lambda_C \equiv \lambda_p\frac{\sigma z_C}{1-z_C}, \quad \psi_a \equiv \lambda_p(1+\chi), \quad \lambda_p \equiv \frac{(1-\theta_p)(1-\theta_p\beta)}{\theta_p}, \quad (34)$$

[4] The borrower debt-elastic closing device affects the bond-pricing block; it is not appended directly to the IS curve.

where Λ_C scales the credit-induced marginal-cost offset and ψ_a is the pass-through from productivity.[5]

3.3 Propagation Through the Credit Distribution

Define

$$s_j \equiv \frac{\bar{\Delta}\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \qquad e_j \equiv \frac{\kappa\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \tag{35}$$

where s_j is the share of bank j's steady-state fundamental value attributable to lending operations and e_j captures the sensitivity to the inherited bond portfolio. Let $\mathbf{S} \equiv \mathrm{diag}(s_1, \ldots, s_n)$ and $\mathbf{E} \equiv \mathrm{diag}(e_1, \ldots, e_n)$. The two coefficients separate two distinct components of bank value: s_j measures how much of bank j's steady-state value is generated *currently* by the spread on lending, while e_j measures how much is *inherited* from past lending decisions through the bond portfolio that the bank carries on its balance sheet. The first will load on contemporaneous spread and price shocks; the second will load on the inherited portfolio and is what gives the credit distribution its endogenous dynamics.

Substituting the linearized fundamental values into the leverage constraints gives:

$$\hat{\mathbf{b}}_t^{FI} := \underbrace{\left[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}\right]^{-1}}_{\text{damped credit multiplier}} \Big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\,\hat{q}_t + \boldsymbol{\alpha}_\Delta\,\hat{\Delta}_t + \underbrace{\left(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}\right)}_{\text{augmented legacy}}\,\hat{\mathbf{b}}_{t-1}^{FI}\Big], \tag{36}$$

where

$$\boldsymbol{\alpha}_q \equiv \mathcal{A}(\mathbf{S}+\mathbf{E})\mathbf{1} - \mathbf{1}, \qquad \boldsymbol{\alpha}_\Delta \equiv \mathcal{A}\mathbf{S}\mathbf{1}. \tag{37}$$

Equation (36) reads as follows. The bracketed term collects everything that pushes lending up at the bank level: capital injections $\hat{\boldsymbol{\phi}}_t$, valuation effects from asset prices and spreads ($\boldsymbol{\alpha}_q$ and $\boldsymbol{\alpha}_\Delta$), and inherited portfolio positions through the legacy term. The leading inverse is the *damped credit multiplier*: $\mathcal{A}\mathbf{S}$ captures the network feedback by which a one-dollar increase in lending at bank j raises every other bank's market value through the cross-holdings matrix and thereby relaxes their leverage constraints; the $(1+\eta_B)$ on the diagonal keeps that loop convergent and bounds how quickly portfolios can adjust. Without the network ($\mathcal{A} = \mathbf{I}$), the inverse degenerates to a scalar $1/(1+\eta_B - s_j)$ and the credit distribution becomes a mechanical sum of bank-level shocks. The coefficients $\boldsymbol{\alpha}_q$ and $\boldsymbol{\alpha}_\Delta$ play the analogous role for asset-price and spread shocks: $\mathcal{A}$ converts bank-level fundamentals into market values before leverage constraints bind, so the asset-price loading depends on both current and legacy exposures while the spread loading depends only on current spread income.

The inverse exists under the matrix stability condition $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$, which

[5] The parameter θ_p is the Calvo non-adjustment probability and is distinct from the bank portfolio adjustment cost η_B introduced in Section 2.1.2.

rules out an explosive contemporaneous loop from lending to valuation and back to lending.

The persistence of the credit distribution is governed by

$$\mathbf{P} \equiv \big[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}\big]^{-1}\big(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}\big), \tag{38}$$

the matrix that maps last period's credit distribution into this period's. $\mathbf{P}$ is what makes the cross section an *endogenous macroeconomic state*: a monetary policy shock that today shifts lending toward one set of banks leaves a footprint on $\hat{\mathbf{b}}^{FI}_{t-1}$, and that footprint keeps influencing $\hat{\mathbf{b}}^{FI}_{t}$ through $\mathbf{P}$ in subsequent periods. The two terms in $\mathcal{A}\mathbf{E}+\eta_B\mathbf{I}$ correspond to two distinct sources of memory: $\mathcal{A}\mathbf{E}$ is the *economic* channel by which inherited bond portfolios continue to generate market value through the network, and $\eta_B\mathbf{I}$ is the *frictional* channel by which adjustment costs anchor portfolios to their past. Section 4 shows that this endogenous state is the mechanism behind the financial-memory effect: past accommodation reshapes the credit distribution and thereby alters the transmission of future policy.

3.4 From Bank-Level Lending to Aggregate Capacity

We now take the key aggregation step. Each bank's leverage constraint determines how much that bank lends individually; we want a single equation describing how much the banking sector lends in aggregate. The natural weighting uses steady-state lending shares $\omega_i \equiv \bar{q}\bar{B}_i / \sum_j \bar{q}\bar{B}_j$, but an asymmetry emerges along the way: the leverage constraint of bank i binds against its *market value*, which through the network is a weighted sum of *all* banks' fundamental values. As a result, when we weight constraint by constraint, the loading on bank j's lending is not its own market share ω_j, but the share of aggregate market value that bank j ultimately drives. Carrying out the algebra:

$$\Theta^{\text{net}}_t := \hat{\phi}^{agg}_t + \bar{\mu}\,\hat{\Delta}_t + \bar{\alpha}\,\hat{q}_t + \mathbf{\Lambda}^{(0)}\hat{\mathbf{b}}^{FI}_t + \mathbf{\Lambda}^{(1)}\hat{\mathbf{b}}^{FI}_{t-1} - \eta_B\big(\hat{b}^{FI}_t - \hat{b}^{FI}_{t-1}\big), \tag{39}$$

where $\hat{\phi}^{agg}_t \equiv \sum_i \omega_i \hat{\phi}_{i,t}$ and $\hat{b}^{FI}_t \equiv \sum_i \omega_i \hat{b}_{i,t}$. Equation (39) collapses the entire bank-level credit distribution into a single scalar—net financial capacity—that the rest of the macroeconomy interacts with. The structure of the aggregation coefficients is governed by a single object, the log-linear *effective network weight*:

$$\tilde{\omega}^{\ell}_j \equiv \sum_{i\in\mathcal{N}} \omega_i\,\mathcal{A}_{ij}. \tag{40}$$

To unpack what $\tilde{\omega}^{\ell}_j$ means, consider the following thought experiment. Suppose bank j alone receives a small positive shock to its fundamental value—say, a one-percent rise in its lending margin. Through the elasticity matrix $\mathcal{A}$, this shock raises the market value

of every bank that holds j's equity, directly or indirectly, and therefore relaxes their leverage constraints by a heterogeneous amount. Weighting each bank's response by its contribution ω_i to aggregate lending and summing gives exactly $\tilde{\omega}_j^\ell$. Two implications follow.

First, $\tilde{\omega}_j^\ell$ generally differs from the market share ω_j. A bank that is heavily held by other banks—especially by banks that themselves have large ω_i—has a $\tilde{\omega}_j^\ell$ larger than ω_j, even if its own balance sheet is modest. Conversely, a peripheral bank's lending matters only in proportion to its own size, so its effective weight is close to its market share. The wedge $\tilde{\omega}_j^\ell - \omega_j$ is therefore a measure of how systemically connected bank j is in the sense relevant for monetary transmission, and it will drive the counterfactual exercises in Section 4.

Second, every coefficient in equation (39) factors as an effective network weight times a source of value exposure:

$$\bar{\mu} = \sum_j \tilde{\omega}_j^\ell \, s_j, \quad \bar{\alpha} = \sum_j \tilde{\omega}_j^\ell \, (s_j + e_j), \quad \Lambda_j^{(0)} = \tilde{\omega}_j^\ell \, s_j, \quad \Lambda_j^{(1)} = \tilde{\omega}_j^\ell \, e_j. \tag{41}$$

Spread shocks and current bond-price movements load on the spread-income exposure s_j, asset-price effects also load on the legacy exposure e_j, and inherited credit positions load on e_j alone. In every case, the loading on bank j is its effective network weight multiplied by the relevant component of j's steady-state value.

The level dependency matrix $\mathbf{A}$ remains the structural network object; the effective weight $\tilde{\omega}_j^\ell$ is its log-linear aggregation counterpart, capturing how a unit log change in bank j's fundamental value contributes to aggregate financial capacity after accounting for steady-state valuation shares.

3.5 The Credit Spread

The credit spread is the discounted excess return on bank lending over deposits, evaluated under the nominal stochastic discount factor:

$$\Delta_t \equiv \mathbb{E}_t[M_{t,t+1}^N (R_{t+1}^b - R_t^s)]. \tag{42}$$

The nominal SDF is the relevant pricing kernel because both returns are nominal cash flows; this is the profitability margin that enters bank fundamental value through s_j in Section 3.3. Let $\mathcal{R} \equiv \beta/\beta_B > 1$, so $\bar{\Delta} = \mathcal{R} - 1$. Log-linearizing the discounted spread and using the saver Euler equation $\mathbb{E}_t[M_{t,t+1}^N R_t^s] = 1$ gives:

$$\hat{\Delta}_t = \frac{\mathcal{R}}{\mathcal{R} - 1} \big(\mathbb{E}_t \hat{r}_{t+1}^b - \hat{r}_t^s \big). \tag{43}$$

The spread rises when expected long-bond returns increase relative to the deposit rate, scaled by the steady-state excess-return margin. The long bond return linearizes as:

$$\hat{r}^b_{t+1} \approx \beta_B \kappa \, \hat{q}_{t+1} - \hat{q}_t, \tag{44}$$

where $\beta_B \kappa = \kappa \bar{q}/(1 + \kappa \bar{q})$ under the steady-state borrower pricing condition. A higher price today lowers the holding-period return, while a higher expected price tomorrow raises the continuation value of the long bond.

3.6 The Four-Equation System

The model collapses to four equations linking the macroeconomic state $\{\hat{y}_t, \hat{\pi}_t, \Theta^{\text{net}}_t, \hat{\mathbf{b}}^{FI}_t\}$:

- the IS curve (32) carries the standard intertemporal substitution channel *plus* a financial channel that operates through changes in Θ^{net}_t and $\widehat{qe}_t$;
- the NKPC (33) adds to the textbook output channel a marginal-cost effect of the *level* of financial capacity, because a credit-driven shift in the borrower–saver consumption split changes saver labor supply;
- the credit-distribution law (36) determines which banks lend, evolving through the damped credit multiplier $\left[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}\right]^{-1}$ and giving the cross section the status of an endogenous state;
- the network aggregation (39) collapses the cross section into the scalar Θ^{net}_t via effective network weights $\tilde{\omega}^{\ell}_j$.

The system is closed by the Taylor rule for $\hat{r}^s_t$, the QE rule for $\widehat{qe}_t$, the spread equation (43), the bond-return equation (44), and exogenous processes for $\hat{a}_t$ and $\hat{\boldsymbol{\phi}}_t$. The role of the dependency network $\mathbf{A}$, summarized through the effective weights $\tilde{\omega}^{\ell}_j$, enters *both* as a level shifter in the IS–NKPC block and as a propagation matrix in the credit distribution. The next section uses this structure to characterize how monetary policy is transmitted, and how that transmission depends on the alignment between $\tilde{\omega}^{\ell}_j$ and ω_j.

4 The Financial Reallocation Channel of Monetary Policy

Monetary policy reallocates credit across banks. The bank-level mechanism of Section 2 pins down the responsive margin (which banks expand lending most), and the four-equation system of Section 3 pins down the aggregation margin (whose lending matters most for the banking system's overall capacity to fund borrowers). The interaction between the two margins is what we call the *financial reallocation channel* of monetary

policy: a single monetary shock shifts the cross-section of bank credit, and the network determines how much of that cross-sectional shift survives into a movement in aggregate financial capacity.

Two bank-level indices describe the channel. The dependency index A_{ii} governs how strongly bank i's own lending feeds back into its own market value and so relaxes its leverage constraint; it is the bank's *responsiveness* to monetary policy. The effective network weight $\tilde{\omega}_j^\ell$ governs how much a movement in bank j's fundamental value moves the market-value base of the entire banking sector; in plain English we will refer to $\tilde{\omega}_j^\ell$ as bank j's *systemic importance* in this paper's sense—how much its lending spills over onto other banks' ability to lend.

Whether monetary policy generates a large or a small movement in aggregate financial capacity depends on the *alignment* between responsiveness and systemic importance. When the most responsive banks are also systemically important, the cross-sectional reallocation delivers a large aggregate-capacity effect; when the two features are misaligned, monetary expansion produces a sizable reshuffle of credit across banks but only a modest aggregate response. This single alignment statement organizes the rest of the section. We trace its implications for four properties of monetary transmission: the operation of the channel from a bank-level shock to aggregate capacity (Section 4.2), the channel's footprint in the IS curve and the Phillips curve (Section 4.4), the relative size of the output and inflation responses to the *same* monetary shock—the output–inflation *mix* of transmission (Section 4.5.1)—the differential amplification of conventional rate policy and quantitative easing (Section 4.5.2), and the persistence of past policy through inherited bond holdings (Section 4.5.3).

4.1 Bank Responsiveness vs. Systemic Importance

The PE and GE objects answer different economic questions. The dependency index A_{ii} asks how much of bank i's own fundamental value is retained in bank i's own market value after all ownership feedback and value leakage are accounted for. It is therefore the object that governs bank i's lending elasticity in Proposition 2. The market share ω_j asks how much credit bank j directly holds in steady state. Bank j's effective network weight $\tilde{\omega}_j^\ell$, or in plain English its *systemic importance* in this paper's sense, asks a third question: when bank j's fundamental value changes, how much does the market-value base of the entire banking sector move?

The answer is

$$\tilde{\omega}_j^\ell \equiv \sum_i \omega_i \mathcal{A}_{ij}. \tag{45}$$

The term $\mathcal{A}_{ij}$ says how much bank j's log fundamental value changes bank i's log market value; the weight ω_i says how much bank i matters in aggregate lending capacity. A bank

can therefore be highly responsive without being macroeconomically central, and it can be macroeconomically central without being the most responsive bank.

This is where the partial-equilibrium results enter the macroeconomic analysis. PE determines the direction of credit reallocation across banks; GE determines how much that reallocation changes Θ_t^{net}. In a representative-intermediary model, the question "which bank holds the credit" has no aggregate content: only the total quantity matters. Here the same aggregate stock of credit can generate different values of Θ_t^{net} depending on whether the marginal lending is carried by banks with high or low $\tilde{\omega}_j^{\ell}$.

4.2 How the Reallocation Channel Operates

The reallocation channel runs through three stages—from a bank-level shock, to a cross-sectional shift in credit, to a movement in aggregate financial capacity—each of which we now make precise.

Stage 1 (Reallocation). By Propositions 2–3, easing pushes credit toward high-A_{ii} banks. This stage locates marginal credit but does not yet pin down its aggregate weight.

Stage 2 (Aggregation). The network turns the cross-sectional shift into movement in Θ_t^{net} only after weighting by each bank's systemic importance $\tilde{\omega}_j^{\ell}$. Highly responsive banks with low $\tilde{\omega}_j^{\ell}$ move the cross-section without moving aggregate capacity. Section 4.3 states this idea formally as a single cross-sectional covariance.

Stage 3. Household redistribution and the inflation channel. Once aggregate financial capacity moves, the borrower-saver structure delivers the supply-side bite. Borrower consumption is tied to the real value of intermediary- and central-bank-financed debt; given output, a rise in borrower consumption mechanically lowers saver consumption,

$$\hat{c}_{P,t} = \frac{1}{1 - z_C}\hat{y}_t - \frac{z_C}{1 - z_C}\big(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\big). \tag{46}$$

A lower $\hat{c}_{P,t}$ raises the saver's marginal utility of income; through the labor-supply condition this shifts effective labor supply outward and lowers the real marginal cost required to produce a given level of output. The reallocation channel therefore shows up in inflation not only through the level of aggregate demand but also through the level of saver consumption.

4.3 The Alignment Covariance

The alignment statement that opens this section can be made precise. We ask the following question: holding aggregate bank credit fixed, how much does monetary easing change aggregate financial capacity by reallocating credit *across* banks? The answer turns out to be a single cross-sectional covariance, evaluated at steady state.

The derivation is collected in Appendix C.1.

Proposition 6 (Alignment Covariance)**.** *Holding aggregate bank credit fixed, the contribution of cross-sectional credit reallocation to aggregate financial capacity following a monetary shock m is*

$$\frac{d\Theta_t^{realloc}}{dm} = \bar{\kappa}\,\mathrm{Cov}_\omega\big(\xi_i,\, Z_i^m\big), \tag{47}$$

where Z_i^m is bank i's lending semi-elasticity to the monetary shock,[6] *$\xi_i \equiv \tilde{\omega}_i^\ell - \omega_i$ is bank i's capacity wedge—the gap between its effective network weight and its steady-state market share, with a positive wedge identifying banks whose lending matters more for aggregate capacity than their balance-sheet share alone implies—$\bar{\kappa} > 0$ is a positive scaling factor that aggregates effective weights and spread exposure across banks, and the covariance is taken across banks weighted by their steady-state market shares.*

Three features of this expression deserve emphasis.

Sign. The reallocation channel raises aggregate financial capacity if and only if the cross-sectional covariance is positive; that is, if and only if banks with high lending responses also tend to have positive capacity wedges. The sign of the channel is the sign of the alignment.

Strength. A larger covariance produces a larger reallocation contribution to aggregate capacity. Two banking systems with the same total credit, the same average lending response, and the same average capacity wedge can deliver very different reallocation effects if their cross-sectional alignments differ.

Vanishing. The reallocation channel disappears in three economically distinct cases: (i) when all banks have the same lending response, so policy moves no credit; (ii) when all banks have the same capacity wedge, so the wedge cannot covary with anything; and (iii) when the network degenerates and the wedge is uniformly zero, in which case effective weights coincide with market shares and we recover the representative-intermediary limit of Section 4.5.4.

The remainder of this section traces the alignment covariance through the IS curve, the Phillips Curve, the differential between QE and rate policy, and financial memory. In each case the qualitative claim reduces to the same alignment idea: monetary policy moves the macroeconomy through the cross-section of bank credit only when responsiveness and systemic importance covary in the right direction.

[6] Z_i^m is governed by the dependency index A_{ii} and is the source of the cross-sectional reallocation in Proposition 3.

4.4 The Reallocation Channel in the IS Curve and Phillips Curve

The same financial-capacity variable enters the two core macroeconomic equations in different ways. In the IS curve it enters in changes:

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{1-z_C}{\sigma}\big(\hat{r}_t^s - \mathbb{E}_t\hat{\pi}_{t+1}\big) + z_C\bar{b}^{FI}\big(\Theta_t^{\text{net}} - \mathbb{E}_t\Theta_{t+1}^{\text{net}}\big) + z_C\bar{b}^{CB}\big(\widehat{qe}_t - \mathbb{E}_t\widehat{qe}_{t+1}\big). \quad (48)$$

The reason is intertemporal. Borrower consumption rises when current financial capacity is high relative to expected future financial capacity. A temporary easing that raises current bank lending or current QE purchases stimulates current borrower spending. If the same increase is expected to persist one-for-one into the future, the intertemporal impulse to current demand is weaker.

In the Phillips Curve, financial capacity enters in levels:

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\hat{y}_t - \Lambda_C\big(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\big) - \psi_a\hat{a}_t. \quad (49)$$

Inflation depends on today's marginal cost. For a given output level, a higher level of borrower consumption implies lower saver consumption today. Lower saver consumption increases the marginal value of income for the household that supplies labor, which lowers the real wage required to support a given amount of labor. Marginal cost falls today even if the movement in financial capacity is expected to persist. How large the financial-capacity movements feeding these two equations turn out to be is itself governed by the alignment covariance derived in Section 4.3.

We will refer to this gap between the demand-side and supply-side imprints of monetary easing as the *output–inflation composition* of transmission.[7] A textbook financial accelerator says that easing raises credit and therefore aggregate demand. Here the network also determines how much of that credit expansion shows up as a supply-side offset, because credit composition—not just credit quantity—moves saver consumption. The strength of the offset relative to the direct output term is

$$\frac{\Lambda_C}{\kappa_y} = \frac{\sigma z_C}{\chi(1-z_C)+\sigma}. \quad (50)$$

The term z_C controls how much credit-financed borrower spending matters in goods-market clearing. The curvature parameter σ controls how strongly saver marginal utility reacts when saver consumption falls. The network does not change these preference parameters; it changes the financial-capacity movement to which these parameters are applied. Equivalently, the same household block can deliver very different output–inflation compositions depending on whose lending the central bank ultimately mobilizes through

[7] Throughout, the "output–inflation mix" refers to the relative size of the two endogenous responses to a single monetary shock, not a Phillips-curve tradeoff faced by a policy maker.

the network.

4.5 Implications of the Reallocation Channel

The alignment covariance and its imprint on the IS curve and the Phillips Curve combine to deliver four downstream results. Each is a distinct prediction of how monetary transmission depends on the banking network. Section 4.5.1 studies the relative size of the output and inflation responses to the same monetary shock. Section 4.5.2 explains why the network amplifies quantitative easing and conventional rate policy by different amounts, conditional on a common balance-sheet multiplier. Section 4.5.3 traces the persistence of past policy through inherited bond holdings. Section 4.5.4 closes the section by characterizing the degenerate-network limit in which the channel vanishes and the representative-intermediary benchmark is recovered.

4.5.1 Output–Inflation Mix

To make precise how the network reshapes the relative size of output and inflation responses to a single monetary shock, substitute the credit distribution law into the aggregation equation. The contemporaneous transmission from bank-level shocks and prices into aggregate capacity is summarized by

$$\boldsymbol{\Gamma}' \equiv \boldsymbol{\Lambda}^{(0)}[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}. \tag{51}$$

The row vector $\boldsymbol{\Gamma}'$ combines the two margins emphasized in Section 4.1. $\boldsymbol{\Lambda}^{(0)}$ identifies where effective network weights are large in aggregation, and the inverse matrix identifies how shocks, spreads, and asset prices propagate through bank-level credit positions. A bank matters most for aggregate capacity when it is both important in aggregation and connected to propagation paths that make its lending response large.

The reduced-form financial capacity equation is:

$$\begin{aligned}\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \boldsymbol{\Gamma}'\hat{\boldsymbol{\phi}}_t + \underbrace{\left(\bar{\mu} + \boldsymbol{\Gamma}'\boldsymbol{\alpha}_\Delta\right)}_{\equiv \mu^*}\hat{\Delta}_t + \underbrace{\left(\bar{\alpha} + \boldsymbol{\Gamma}'\boldsymbol{\alpha}_q\right)}_{\equiv \alpha^*}\hat{q}_t \\ + \underbrace{\left[\boldsymbol{\Lambda}^{(1)} + \boldsymbol{\Gamma}'(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\right]}_{\equiv \boldsymbol{\Psi}'}\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}).\end{aligned} \tag{52}$$

The coefficients μ^* and α^* show how spreads and bond prices move aggregate financial capacity after bank-level responses are network-weighted. The vector $\boldsymbol{\Psi}'$ shows how inherited credit positions enter today's capacity. The adjustment-cost term subtracts rapid aggregate portfolio changes, so a network can contain high effective-weight banks and still generate a muted aggregate response when portfolio adjustment is sluggish.

For a monetary disturbance, define

$$\gamma_t \equiv \left.\frac{\partial \Theta_t^{\text{net}}}{\partial \hat{y}_t}\right|_{\text{monetary shock}}. \tag{53}$$

This derivative measures how much financial capacity moves together with output under the policy experiment being considered. Because the derivative is taken along a particular monetary experiment, γ_t is not a primitive parameter; it is a reduced-form sufficient statistic for how that experiment moves financial capacity together with output. Substituting this co-movement into the Phillips Curve gives an equivalent representation:

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \tilde{\kappa}_t \hat{y}_t + \tilde{u}_t, \qquad \tilde{\kappa}_t \equiv \kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t. \tag{54}$$

The textbook coefficient κ_y measures how inflation responds to output holding the financial redistribution term fixed. The second term captures the marginal-cost relief generated by the financial-capacity movement that accompanies monetary easing. A larger positive co-movement γ_t means that the same output response is accompanied by a larger supply-side credit offset and therefore by a smaller contemporaneous inflation response.

The economic content of γ_t is the alignment statement of Section 4's opening. γ_t is large when easing pushes credit toward banks that are simultaneously highly responsive to monetary disturbances (large semi-elasticities, high A_{ii}) and macroeconomically central in aggregation (high $\tilde{\omega}_j^{\ell}$). It is small or zero when responsive banks have low effective weights, when value dissipation through the network is strong, or when portfolio adjustment costs η_B dampen the cross-sectional response. The proposition below is the formal statement that follows from this logic.

The reduced-form coefficient γ_t is the GE generalization of the alignment covariance. For a pure credit-spread experiment with aggregate bank credit held fixed, γ_t is proportional to $\bar{\kappa}\,\mathrm{Cov}_\omega(\xi_i, Z_i^m)$ from Proposition 6. For richer monetary experiments, γ_t also picks up the aggregate-credit term and the joint response of asset prices and the SDF; the covariance form remains the dominant cross-sectional driver because the other terms are common to all banks and do not depend on the alignment.

Proposition 7 (Financial Reallocation and the Output–Inflation Mix)**.** *For monetary shocks with $\gamma_t > 0$, an increase in network financial capacity creates a marginal-cost offset to the inflationary effect of higher output. Relative to a no-network economy, monetary transmission generates a larger real response per unit of inflation response if and only if $\gamma_t^{net} > \gamma_t^{no\,net}$. This condition is more likely when banks with high response elasticities also have large effective network weights in aggregation.*

The condition in the proposition is the GE version of the PE reallocation results. Proposition 2 identifies the banks that respond most to easing, and Proposition 3 tells us

whether credit moves toward them. The composition result asks a second question: do those high-response banks also have large aggregate-capacity effects? When the answer is yes, the same credit expansion generates a larger increase in Θ_t^{net}, a stronger redistribution from savers to borrowers, and a larger marginal-cost offset. Monetary policy then produces more real activity relative to inflation.

The condition can fail for economically informative reasons. If value dissipation is strong, if the banks expanding most have low effective weights, or if η_B makes the portfolio response sluggish, the network may add little to γ_t. A symmetric network illustrates the same point. Proposition 5 implies that a vertex-transitive network can amplify all banks uniformly without generating credit reallocation. In that case the network may change the level of financial transmission, but the output–inflation *mix* channel is muted. A heterogeneous network matters for the mix only when it moves credit across banks with different effective network weights.

4.5.2 Differential Amplification: Rate Policy vs. Quantitative Easing

The same PE-GE logic explains why different monetary policy *instruments* can receive different amounts of network amplification, even with no asymmetry between expansion and contraction. The partial-equilibrium propositions apply to the component of policy that changes bank balance sheets, spreads, and bond portfolios; they do not apply to the component that reaches households directly through the Euler equation. The network therefore amplifies the bank-balance-sheet share of a policy instrument, not the entire policy instrument mechanically.

Conventional interest rate policy has both components. A cut in $\hat{r}_t^s$ raises saver demand through the Euler equation. This is the standard New Keynesian channel and does not depend on which banks hold credit. The same rate cut also changes spreads and long-bond prices, which affects bank lending and therefore Θ_t^{net}. Only this second component is filtered through $\tilde{\omega}_j^\ell$ and the credit multiplier.

QE is more concentrated in the balance-sheet component. A purchase of long bonds changes the quantity of bonds held outside the central bank, raises bond prices, and changes the value of portfolios inherited by new bankers. These movements operate through the coefficients $\boldsymbol{\alpha}_q$, $\mathbf{E}$, and the aggregation weights in the financial block. QE therefore loads more heavily on the same feedback-and-reallocation mechanism identified by Proposition 2 and Proposition 3.

Proposition 8 (Conditional Differential Amplification Across Instruments)**.** *Suppose the network multiplier raises the balance-sheet component of both policy instruments by a common factor greater than one, and suppose the interest-rate instrument has a positive direct Euler-equation component that is invariant to the network. Then the output amplification ratio for QE exceeds the output amplification ratio for conventional interest-rate policy.*

The comparison is about ratios, not only levels. If the network increases the balance-sheet response by the same proportional amount for both instruments, the instrument with the larger balance-sheet share receives the larger overall network amplification. QE has a larger such share because its transmission is routed through bond markets and intermediary balance sheets. Rate policy has a household bypass, so part of its output effect is present even when the network is shut down.

This also changes the output–inflation *mix* of the two instruments in the sense of Section 4.4. A policy tool that delivers more of its output response through Θ_t^{net} also delivers more of the marginal-cost offset described above. QE therefore generates a larger real response with a comparatively muted contemporaneous inflation response whenever its balance-sheet channel is strong. The claim does not require QE to dominate rate policy in every calibration; it requires only that the network-amplified share of QE transmission be larger than the network-amplified share of rate policy.

4.5.3 Memory through Inherited Bond Holdings

The financial reallocation channel also has memory. Monetary policy changes the cross-sectional allocation of bank credit, and that allocation persists because banks enter subsequent periods with legacy bond portfolios. These inherited portfolios enter the equity transfer to new bankers and affect current market values, current leverage constraints, and current network financial capacity. The log-linearized system around the steady state—which we refer to below as the *first-order system*—summarizes this persistence through the matrix

$$\mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}). \tag{55}$$

When $\rho(\mathbf{P})$ is close to one, credit reallocations fade slowly. Iterating the credit law forward shows that the effect of inherited positions decays with powers of $\mathbf{P}$. A period of easing that tilts credit toward a particular set of banks therefore leaves a trace in the next period's financial capacity. The term $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ is the macroeconomic expression of this inherited balance-sheet channel. It shows that the same aggregate stock of bonds can support different financial capacity depending on which banks hold it.

In the first-order system, $\mathcal{A}$ and $\tilde{\omega}^{\ell}$ are fixed around the steady state. Financial memory therefore operates through inherited wedges rather than through time-varying local coefficients. From (52), inherited credit positions enter through $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$. Substituting this term into the NKPC shifts the inflation residual:

$$-\Lambda_C \bar{b}^{FI} \mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}.$$

Past policy changes the financial position from which the next policy shock is transmitted, even when the local coefficients are fixed.

Proposition 9 (Financial Memory Through Inherited Bond Holdings)**.** *In the first-order system with fixed $\mathcal{A}$ and $\tilde{\omega}^{\ell}$, the inherited cross-section of bank bond holdings affects current network financial capacity through $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ and evolves according to the persistence matrix $\mathbf{P}$. The persistence of this inherited financial wedge is increasing in $\rho(\mathbf{P})$.*

This is a form of financial memory. If past monetary expansion leaves the banking sector tilted toward high-effective-weight institutions, the economy enters the next period with a larger financial-capacity wedge; if past tightening shifts credit away from those institutions, the wedge moves in the opposite direction. The linear benchmark therefore separates two layers of effects. History reshapes the inherited cross-section of bond holdings $\hat{\mathbf{b}}_{t-1}^{FI}$, while the steady-state network objects—market shares ω_j and effective weights $\tilde{\omega}_j^{\ell}$—are held fixed. Allowing those objects themselves to drift with the state would require a fully nonlinear or higher-order approximation; we treat that as a question for future work.

4.5.4 The Representative-Intermediary Limit

A final natural sanity check is to ask when the representative-intermediary New Keynesian benchmark is recoverable from our model. The degenerate network case provides exactly that limit and clarifies what the network adds. The borrower-saver structure already creates a supply-side credit wedge: when borrower consumption rises and saver consumption falls, marginal cost declines. That mechanism does not require a network. What the network adds is composition dependence—making the size of the wedge depend on *which* banks supply the marginal credit, not just on *how much* credit is supplied in aggregate.

Proposition 10 (Degenerate Network Limit)**.** *When $\mathbf{C} = \mathbf{0}$, the level dependency matrix is $\mathbf{A} = \mathbf{I}$ and the log-elasticity matrix is also $\mathcal{A} = \mathbf{I}$. The effective weights collapse to market shares, $\tilde{\omega}_j^{\ell} = \omega_j$, and the credit multiplier reduces to $[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}$. If banks are ex ante identical, the credit distribution is irrelevant for aggregation and the model collapses to a representative-intermediary New Keynesian system augmented by a scalar financial capacity state.*

When $\mathcal{A} = \mathbf{I}$, a bank's fundamental value affects only its own market value. There is no distinction between the bank that holds credit and the bank whose value matters for the rest of the system. With identical banks, reallocating a fixed amount of credit across institutions does not change aggregate financial capacity. The cross-section can be collapsed without losing the macroeconomic mechanism. Equivalently, in this limit the capacity wedge ξ_i vanishes for every bank, so the alignment covariance of Proposition 6 is identically zero regardless of how heterogeneous bank-level lending responses are.

When $\mathcal{A} \neq \mathbf{I}$, that collapse is no longer valid. A credit shift toward a bank with a high market share need not have the same macroeconomic effect as a shift toward a

bank with a high effective weight. The network contribution is therefore composition dependence: the aggregate effect of monetary policy depends on the identity of the banks that expand, because the network determines how their lending maps into system-wide financial capacity.

This is the sense in which the effective network weight is the central object of the paper. It connects micro-level network positions to the aggregate IS curve, the marginal-cost channel in the NKPC, the output–inflation composition of policy, and the relative amplification of QE and rate policy. The paper's mechanism is not that networks always amplify monetary shocks. It is that networks decide *which* credit expansions matter most for aggregate outcomes, and *by how much* the resulting movement in financial capacity feeds back into the borrower-saver redistribution that determines marginal cost.

5 Conclusion

This paper studies how financial networks change the composition of monetary transmission. In a representative-intermediary model, only the aggregate quantity of credit matters. Once banks are connected through balance-sheet linkages, the identity of the banks supplying marginal credit also matters. Monetary policy then affects the economy not only by changing total credit, but by reallocating credit across banks whose network positions imply different aggregate consequences.

The central object is the effective network weight. It links bank-level lending responses to aggregate financial capacity and separates two questions that are identical in a representative-intermediary model: which banks expand after policy easing, and whose expansion matters most for output and inflation. This distinction generates a financial reallocation channel, helps explain why balance-sheet policies can be amplified differently from interest-rate policy, and creates financial memory through inherited cross-sectional bond holdings.

Several extensions are natural. Empirically, the model points to the measurement of effective network weights and their relationship with observed monetary-policy responses across banks. Theoretically, allowing market shares, network weights, or interbank exposures to evolve away from the steady state would turn inherited financial wedges into richer state dependence. These extensions would sharpen the link between the financial architecture of the banking system and the changing composition of monetary transmission.

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A Model Appendix: Settings, First-Order Conditions, and Reduction

This appendix states the complete model environment, first-order conditions, market-clearing conditions, and the algebraic reduction to the four-equation system used in the paper. Throughout, hatted variables denote log deviations from the non-stochastic steady state, $\hat{x}_t \equiv \log(x_t/\bar{x})$.

The level dependency matrix and its log-linear counterpart are distinct:

$$\mathbf{A}(\tau) = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}, \qquad \mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}. \tag{56}$$

The matrix $\mathbf{A}$ maps fundamental value levels into outside market-value levels, while $\mathcal{A}$ maps log deviations:

$$\mathbf{v}_t = \mathbf{A}\mathbf{x}_t, \qquad \hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\hat{x}_{j,t}. \tag{57}$$

A.1 Timing, Households, and Asset Pricing

The economy is in discrete time. At the beginning of period t, the previous generation of bankers exits and distributes dividends. New bankers enter and receive equity transfers. Households choose consumption, labor, saving, and borrowing. Banks choose long-bond and reserve portfolios subject to leverage. Firms produce and Calvo-adjusting retailers reset prices. The central bank sets the short rate and its bond portfolio according to policy rules. Returns are realized between t and $t+1$.

The long bond bought at t pays one unit at $t+1$, κ at $t+2$, and so on. Its gross nominal holding-period return is

$$R^b_{t+1} = \frac{1+\kappa Q_{t+1}}{Q_t}. \tag{58}$$

Patient household. The patient household consumes $C_{P,t}$, supplies labor L_t, saves through nominal deposits, owns production-firm equity, and owns outside equity claims on financial intermediaries. Preferences are

$$\mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[\frac{C_{P,t+j}^{1-\sigma} - 1}{1-\sigma} - \psi \frac{L_{t+j}^{1+\chi}}{1+\chi} \right]. \tag{59}$$

The nominal budget constraint is

$$\begin{aligned} P_t C_{P,t} + \sum_i S_{i,t} + \sum_i P_t X^{FI}_{i,t} + P_t X^b_t \leq W_t L_t + \sum_i R^s_{t-1} S_{i,t-1} \\ + P_t D^F_t + P_t D^{P,FI}_t + P_t T_t. \end{aligned} \tag{60}$$

Let Λ_t^P denote the multiplier on the nominal budget constraint. The consumption and labor first-order conditions are

$$C_{P,t}^{-\sigma} = \Lambda_t^P P_t, \tag{61}$$

$$\psi L_t^\chi = \Lambda_t^P W_t. \tag{62}$$

Combining them and defining $w_t = W_t/P_t$ gives

$$\psi L_t^\chi = w_t C_{P,t}^{-\sigma}. \tag{63}$$

The deposit FOC is

$$\Lambda_t^P = \beta \mathbb{E}_t[\Lambda_{t+1}^P R_t^s]. \tag{64}$$

Using the consumption FOC,

$$1 = \mathbb{E}_t \left[\beta \left(\frac{C_{P,t+1}}{C_{P,t}} \right)^{-\sigma} \frac{R_t^s}{\Pi_{t+1}} \right]. \tag{65}$$

The nominal stochastic discount factor is

$$M_{t,t+1}^N = \beta \left(\frac{C_{P,t+1}}{C_{P,t}} \right)^{-\sigma} \frac{1}{\Pi_{t+1}}. \tag{66}$$

Impatient household. The impatient household consumes $C_{B,t}$, issues long-term bonds, and has discount factor $\beta_B < \beta$. Preferences are

$$\mathbb{E}_t \sum_{j=0}^{\infty} \beta_B^j \frac{C_{B,t+j}^{1-\sigma} - 1}{1-\sigma}. \tag{67}$$

The nominal budget constraint is

$$P_t C_{B,t} + B_{t-1}^C \leq Q_t (B_t^C - \kappa B_{t-1}^C) + P_t X_t^b. \tag{68}$$

The transfer rule is

$$P_t X_t^b = (1 + \kappa Q_t) B_{t-1}^C. \tag{69}$$

Substituting (69) into (68) gives

$$P_t C_{B,t} = Q_t B_t^C. \tag{70}$$

The borrower optimality condition with a debt-elastic closing device is

$$1 = \beta_B \left(\frac{C_{B,t+1}}{C_{B,t}} \right)^{-\sigma} \frac{R_{t+1}^b}{\Pi_{t+1}} - \psi_b \left(\frac{b_{C,t}}{\bar{b}_C} - 1 \right). \tag{71}$$

A.2 Financial Intermediaries

There are n banks indexed by $i \in \mathcal{N} = \{1, \ldots, n\}$. The cross-holdings matrix $\mathbf{C}$ has elements $C_{ij} \in [0, 1)$, where C_{ij} is the fraction of bank j's equity held by bank i. The restrictions are

$$C_{ii} = 0, \qquad \hat{C}_{ii} \equiv 1 - \sum_j C_{ji} \in (0, 1], \qquad \rho(\mathbf{C}) < 1. \tag{72}$$

Let $\hat{\mathbf{C}} = \mathrm{diag}(\hat{C}_{11}, \ldots, \hat{C}_{nn})$. The interbank dividend-chain friction is $\tau \in [0, 1)$: each ownership link transmits only $(1 - \tau)$ of the value it receives.

Network valuation. Fundamental value is

$$x_{i,t} = \mathbb{E}_t[M^N_{t,t+1} D^{FI}_{i,t+1}] = \Delta_t Q_t B_{i,t} + \bar{x}_{i,t}, \tag{73}$$

where

$$\Delta_t \equiv \mathbb{E}_t[M^N_{t,t+1}(R^b_{t+1} - R^s_t)]. \tag{74}$$

Total equity values satisfy

$$\mathbf{V}_t = \mathbf{x}_t + (1 - \tau)\mathbf{C}\mathbf{V}_t, \qquad \mathbf{V}_t = [\mathbf{I} - (1 - \tau)\mathbf{C}]^{-1}\mathbf{x}_t. \tag{75}$$

Outside market value is

$$\mathbf{v}_t = \hat{\mathbf{C}}\mathbf{V}_t = \mathbf{A}(\tau)\mathbf{x}_t, \qquad \mathbf{A}(\tau) \equiv \hat{\mathbf{C}}[\mathbf{I} - (1 - \tau)\mathbf{C}]^{-1}. \tag{76}$$

Its log-linearization is

$$\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij} \hat{x}_{j,t}, \qquad \mathcal{A}_{ij} \equiv \frac{A_{ij}\bar{x}_j}{\bar{v}_i}. \tag{77}$$

Balance sheet, transfers, and dividends. The bank balance sheet is

$$Q_t B_{i,t} + RE_{i,t} = S_{i,t} + P_t X^{FI}_{i,t}. \tag{78}$$

Equity transfers are

$$P_t X^{FI}_{i,t} = P_t \bar{X}^{FI}_i + \kappa Q_t B_{i,t-1}. \tag{79}$$

The fundamental dividend paid by the exiting generation is

$$\begin{aligned} P_t D^{FI}_{i,t} = {} & (R^b_t - R^s_{t-1}) Q_{t-1} B_{i,t-1} + (R^{re}_{t-1} - R^s_{t-1}) RE_{i,t-1} \\ & + R^s_{t-1} P_t X^{FI}_{i,t-1}. \end{aligned} \tag{80}$$

Network dividend distribution is

$$\tilde{\mathbf{D}}_t^{FI} = [\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}\mathbf{D}_t^{FI}. \tag{81}$$

The patient household receives

$$D_t^{P,FI} = \mathbf{1}'\mathbf{A}\mathbf{D}_t^{FI}. \tag{82}$$

Bank problem and KKT conditions. The leverage constraint is

$$Q_t B_{i,t} \leq \phi_{i,t} v_{i,t} - \eta_B \bar{Q}(B_{i,t} - B_{i,t-1}). \tag{83}$$

Under fixed spillovers,

$$v_{i,t} = A_{ii}(\Delta_t Q_t B_{i,t} + \bar{x}_{i,t}) + \bar{v}_{i,t}, \qquad \bar{v}_{i,t} \equiv \sum_{j \neq i} A_{ij} x_{j,t}. \tag{84}$$

Bank i chooses $B_{i,t} \geq 0$ to maximize $v_{i,t}$ subject to (83). Define

$$g_{i,t} \equiv \phi_{i,t} v_{i,t} - \eta_B \bar{Q}(B_{i,t} - B_{i,t-1}) - Q_t B_{i,t} \geq 0. \tag{85}$$

The Lagrangian is

$$\mathcal{L}_{i,t}^{FI} = v_{i,t} + \Omega_{i,t} g_{i,t} + \nu_{i,t} B_{i,t}. \tag{86}$$

The KKT conditions are

$$\Omega_{i,t} \geq 0, \qquad \nu_{i,t} \geq 0, \tag{87}$$

$$g_{i,t} \geq 0, \qquad B_{i,t} \geq 0, \tag{88}$$

$$\Omega_{i,t} g_{i,t} = 0, \qquad \nu_{i,t} B_{i,t} = 0, \tag{89}$$

$$A_{ii}\Delta_t Q_t + \Omega_{i,t}\left(\phi_{i,t} A_{ii} \Delta_t Q_t - \eta_B \bar{Q} - Q_t\right) + \nu_{i,t} = 0. \tag{90}$$

For an interior bond position, $\nu_{i,t} = 0$, hence

$$\Omega_{i,t}\left[Q_t + \eta_B \bar{Q} - \phi_{i,t} A_{ii} \Delta_t Q_t\right] = A_{ii} \Delta_t Q_t. \tag{91}$$

When $\Delta_t > 0$, the objective is increasing in $B_{i,t}$, so the leverage constraint binds. Solving the binding constraint gives

$$B_{i,t}^* = \frac{\phi_{i,t}(A_{ii}\bar{x}_{i,t} + \bar{v}_{i,t}) + \eta_B \bar{Q} B_{i,t-1}}{Q_t(1 - \phi_{i,t} A_{ii} \Delta_t) + \eta_B \bar{Q}}. \tag{92}$$

The bank-side stability restrictions are

$$1 - \phi_{i,t} A_{ii} \Delta_t > 0, \qquad \rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1. \tag{93}$$

A.3 Firms, Policy, Market Clearing, and Equilibrium

The final good firm aggregates differentiated retail varieties:

$$Y_t = \left(\int_0^1 Y_t(f)^{\frac{\epsilon-1}{\epsilon}} df \right)^{\frac{\epsilon}{\epsilon-1}}, \qquad \epsilon > 1. \tag{94}$$

Profit maximization gives demand and the price index:

$$Y_t(f) = \left(\frac{P_t(f)}{P_t} \right)^{-\epsilon} Y_t, \qquad P_t = \left(\int_0^1 P_t(f)^{1-\epsilon} df \right)^{\frac{1}{1-\epsilon}}. \tag{95}$$

Retailers buy wholesale output at nominal marginal cost $P_{m,t}$. Each period, fraction $1-\theta_p$ resets its price. An adjusting retailer solves

$$\max_{P_t^*} \mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k M_{t,t+k} \left[P_t^* Y_{t+k|t} - P_{m,t+k} Y_{t+k|t} \right], \tag{96}$$

subject to

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon} Y_{t+k}. \tag{97}$$

The reset-price FOC is

$$\mathbb{E}_t \sum_{k=0}^{\infty} \theta_p^k M_{t,t+k} Y_{t+k|t} \left[P_t^* - \frac{\epsilon}{\epsilon-1} P_{m,t+k} \right] = 0. \tag{98}$$

Equivalently,

$$p_t^* = \frac{\epsilon}{\epsilon-1} \frac{F_t}{K_t}, \tag{99}$$

$$F_t = Y_t p_{m,t} + \beta \theta_p \mathbb{E}_t [\Pi_{t+1}^{\epsilon} F_{t+1}], \tag{100}$$

$$K_t = Y_t + \beta \theta_p \mathbb{E}_t [\Pi_{t+1}^{\epsilon-1} K_{t+1}]. \tag{101}$$

The price-index recursion and price-dispersion law are

$$1 = \theta_p \Pi_t^{\epsilon-1} + (1-\theta_p)(p_t^*)^{1-\epsilon}, \tag{102}$$

$$\nu_t^p = \theta_p \Pi_t^{\epsilon} \nu_{t-1}^p + (1-\theta_p)(p_t^*)^{-\epsilon}. \tag{103}$$

Log-linearization around zero inflation gives

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \lambda_p \widehat{mc}_t, \qquad \lambda_p = \frac{(1-\theta_p)(1-\theta_p\beta)}{\theta_p}. \tag{104}$$

Wholesale production is

$$Y_{m,t} = A_t L_t, \qquad \ln A_t = \rho_A \ln A_{t-1} + \varepsilon_{A,t}. \tag{105}$$

Cost minimization gives

$$w_t = p_{m,t} A_t. \tag{106}$$

Aggregate production-firm dividends are

$$D_t^F = Y_t - w_t L_t. \tag{107}$$

Monetary policy is rule-based:

$$\frac{R_t^s}{R^s} = \left(\frac{R_{t-1}^s}{R^s}\right)^{\rho_r} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{\phi_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{\phi_y}\right]^{1-\rho_r} e^{\varepsilon_{r,t}}, \tag{108}$$

$$\frac{QE_t}{QE} = \left(\frac{QE_{t-1}}{QE}\right)^{\rho_q} \left[\left(\frac{\Pi_t}{\Pi^*}\right)^{-\lambda_\pi} \left(\frac{Y_t}{Y_t^*}\right)^{-\lambda_y}\right]^{1-\rho_q} e^{\varepsilon_{q,t}}. \tag{109}$$

The central bank balance sheet, real QE definition, interest on reserves, and surplus transfer are

$$Q_t B_t^{cb} = RE_t, \tag{110}$$

$$QE_t \equiv \frac{Q_t B_t^{cb}}{P_t} = q_t b_t^{cb}, \tag{111}$$

$$R_t^{re} = R_t^s, \tag{112}$$

$$P_t T_t = R_t^b Q_{t-1} B_{t-1}^{cb} - R_{t-1}^{re} RE_{t-1}. \tag{113}$$

Market clearing requires

$$B_t^C = \sum_i B_{i,t} + B_t^{cb} = B_t^{FI} + B_t^{cb}, \tag{114}$$

$$RE_t = \sum_i RE_{i,t} = RE_t^{FI}, \tag{115}$$

$$S_t = \sum_i S_{i,t} = S_t^{FI}, \tag{116}$$

$$L_t^s = L_t^d = L_t, \tag{117}$$

$$Y_t = C_{P,t} + C_{B,t}. \tag{118}$$

Using (70), the aggregate resource constraint is

$$Y_t = C_{P,t} + q_t b_t^C. \tag{119}$$

Aggregate production with price dispersion is

$$Y_t \nu_t^p = A_t L_t. \tag{120}$$

An equilibrium is a sequence of allocations, prices, bank values, and policy variables such that households optimize, banks maximize market value subject to leverage, firms maximize profits, the central bank follows the policy rules, values and dividends satisfy the network equations, and all markets clear.

A.4 Step-by-Step Reduction to the Four-Equation System

A.4.1 Aggregate demand

Goods clearing log-linearizes to

$$\hat{y}_t = (1 - z_C)\hat{c}_{P,t} + z_C \hat{c}_{B,t}, \qquad z_C \equiv \frac{\bar{C}_B}{\bar{Y}}. \tag{121}$$

The patient Euler equation log-linearizes to

$$\hat{c}_{P,t} = \mathbb{E}_t \hat{c}_{P,t+1} - \frac{1}{\sigma}\left(\hat{r}_t^s - \mathbb{E}_t \hat{\pi}_{t+1}\right). \tag{122}$$

In real terms, (70) gives

$$C_{B,t} = q_t B_t^C, \qquad \hat{c}_{B,t} = \hat{q}_t + \hat{b}_t^C. \tag{123}$$

Bond clearing log-linearizes to

$$\hat{b}_t^C = \bar{b}^{FI}\hat{b}_t^{FI} + \bar{b}^{CB}\hat{b}_t^{CB}, \qquad \bar{b}^{FI} + \bar{b}^{CB} = 1. \tag{124}$$

Define

$$\Theta_t^{\text{net}} \equiv \hat{q}_t + \hat{b}_t^{FI}, \qquad \widehat{qe}_t \equiv \hat{q}_t + \hat{b}_t^{CB}. \tag{125}$$

Then

$$\begin{aligned}
\hat{c}_{B,t} &= \hat{q}_t + \bar{b}^{FI}\hat{b}_t^{FI} + \bar{b}^{CB}\hat{b}_t^{CB} \\
&= \bar{b}^{FI}(\hat{q}_t + \hat{b}_t^{FI}) + \bar{b}^{CB}(\hat{q}_t + \hat{b}_t^{CB}) \\
&= \bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t.
\end{aligned} \tag{126}$$

Write goods clearing at t and $t+1$:

$$\hat{y}_t = (1-z_C)\hat{c}_{P,t} + z_C\left[\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right], \tag{127}$$

$$\mathbb{E}_t\hat{y}_{t+1} = (1-z_C)\mathbb{E}_t\hat{c}_{P,t+1} + z_C\left[\bar{b}^{FI}\mathbb{E}_t\Theta_{t+1}^{\text{net}} + \bar{b}^{CB}\mathbb{E}_t\widehat{qe}_{t+1}\right]. \tag{128}$$

Subtracting (128) from (127) and applying (122) gives the IS curve:

$$\hat{y}_t = \mathbb{E}_t\hat{y}_{t+1} - \frac{1-z_C}{\sigma}\left(\hat{r}_t^s - \mathbb{E}_t\hat{\pi}_{t+1}\right) + z_C\bar{b}^{FI}\left(\Theta_t^{\text{net}} - \mathbb{E}_t\Theta_{t+1}^{\text{net}}\right) + z_C\bar{b}^{CB}\left(\widehat{qe}_t - \mathbb{E}_t\widehat{qe}_{t+1}\right). \tag{129}$$

A.4.2 Aggregate supply

Real marginal cost satisfies

$$\widehat{mc}_t = \hat{w}_t - \hat{a}_t. \tag{130}$$

Log-linearizing the labor FOC gives

$$\hat{w}_t = \chi\hat{l}_t + \sigma\hat{c}_{P,t}. \tag{131}$$

Around the zero-inflation steady state, price dispersion is second order, so

$$\hat{l}_t = \hat{y}_t - \hat{a}_t. \tag{132}$$

Goods clearing gives

$$\hat{c}_{P,t} = \frac{1}{1-z_C}\hat{y}_t - \frac{z_C}{1-z_C}\hat{c}_{B,t}. \tag{133}$$

Substituting (131), (132), and (133) into (130),

$$\widehat{mc}_t = \left(\chi + \frac{\sigma}{1-z_C}\right)\hat{y}_t - \frac{\sigma z_C}{1-z_C}\hat{c}_{B,t} - (1+\chi)\hat{a}_t. \tag{134}$$

Using (126),

$$\widehat{mc}_t = \left(\chi + \frac{\sigma}{1-z_C}\right)\hat{y}_t - \frac{\sigma z_C}{1-z_C}\left(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right) - (1+\chi)\hat{a}_t. \tag{135}$$

Substituting into the linearized Calvo equation (104) yields

$$\hat{\pi}_t = \beta\mathbb{E}_t\hat{\pi}_{t+1} + \kappa_y\hat{y}_t - \Lambda_C\left(\bar{b}^{FI}\Theta_t^{\text{net}} + \bar{b}^{CB}\widehat{qe}_t\right) - \psi_a\hat{a}_t, \tag{136}$$

where

$$\kappa_y \equiv \lambda_p\left(\chi + \frac{\sigma}{1-z_C}\right), \qquad \Lambda_C \equiv \lambda_p\frac{\sigma z_C}{1-z_C}, \qquad \psi_a \equiv \lambda_p(1+\chi). \tag{137}$$

A.4.3 Credit distribution

At steady state, the adjustment term in (83) is zero and $\bar{Q}\bar{B}_i = \bar{\phi}_i \bar{v}_i$. The binding leverage constraint log-linearizes to

$$\hat{q}_t + \hat{b}_{i,t} + \eta_B(\hat{b}_{i,t} - \hat{b}_{i,t-1}) = \hat{\phi}_{i,t} + \hat{v}_{i,t}. \tag{138}$$

The bank's fundamental value can be written as

$$x_{j,t} = \Delta_t q_t B_{j,t} + \bar{x}_{j,t}, \tag{139}$$

and the transfer rule contains inherited long-bond value:

$$X^{FI}_{j,t} = \bar{X}^{FI}_j + \kappa q_t B_{j,t-1}. \tag{140}$$

The saver Euler equation prices the risk-free component, so the first-order movement in $x_{j,t}$ is

$$\hat{x}_{j,t} = s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1}), \tag{141}$$

where

$$s_j \equiv \frac{\bar{\Delta}\,\bar{q}\,\bar{B}_j}{\bar{x}_j}, \qquad e_j \equiv \frac{\kappa \bar{q}\,\bar{B}_j}{\bar{x}_j}. \tag{142}$$

Using $\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\hat{x}_{j,t}$,

$$\hat{v}_{i,t} = \sum_j \mathcal{A}_{ij}\left[s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1})\right]. \tag{143}$$

Substituting (143) into (138) and collecting contemporaneous credit terms,

$$\begin{aligned}(1+\eta_B)\hat{b}_{i,t} - \sum_j \mathcal{A}_{ij}s_j\hat{b}_{j,t} = \hat{\phi}_{i,t} &+ \left[\sum_j \mathcal{A}_{ij}(s_j + e_j) - 1\right]\hat{q}_t \\ &+ \left[\sum_j \mathcal{A}_{ij}s_j\right]\hat{\Delta}_t + \sum_j \mathcal{A}_{ij}e_j\hat{b}_{j,t-1} + \eta_B\hat{b}_{i,t-1}.\end{aligned} \tag{144}$$

Let $\mathbf{S} = \mathrm{diag}(s_1, \ldots, s_n)$ and $\mathbf{E} = \mathrm{diag}(e_1, \ldots, e_n)$. Define

$$\boldsymbol{\alpha}_q \equiv \mathcal{A}(\mathbf{S} + \mathbf{E})\mathbf{1} - \mathbf{1}, \qquad \boldsymbol{\alpha}_\Delta \equiv \mathcal{A}\mathbf{S}\mathbf{1}. \tag{145}$$

In matrix form,

$$[(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]\,\hat{\mathbf{b}}^{FI}_t = \hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t + (\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\hat{\mathbf{b}}^{FI}_{t-1}. \tag{146}$$

Under $\rho((1+\eta_B)^{-1}\mathcal{A}\mathbf{S}) < 1$, the inverse exists:

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}\left[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t + (\mathcal{A}\mathbf{E} + \eta_B\mathbf{I})\hat{\mathbf{b}}_{t-1}^{FI}\right]. \tag{147}$$

The persistence matrix for inherited credit-distribution modes is

$$\mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}). \tag{148}$$

A.4.4 Network aggregation

Define steady-state lending weights and aggregate bank credit:

$$\omega_i \equiv \frac{\bar{q}\bar{B}_i}{\sum_j \bar{q}\bar{B}_j}, \qquad \hat{b}_t^{FI} \equiv \sum_i \omega_i \hat{b}_{i,t}. \tag{149}$$

Weighting (138) by ω_i and summing yields

$$\Theta_t^{\text{net}} + \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}) = \hat{\phi}_t^{agg} + \hat{v}_t^{agg}, \tag{150}$$

where

$$\hat{\phi}_t^{agg} \equiv \sum_i \omega_i \hat{\phi}_{i,t}, \qquad \hat{v}_t^{agg} \equiv \sum_i \omega_i \hat{v}_{i,t}. \tag{151}$$

Expanding aggregate market value,

$$\begin{aligned} \hat{v}_t^{agg} &= \sum_i \omega_i \sum_j \mathcal{A}_{ij}\hat{x}_{j,t} \\ &= \sum_j \underbrace{\left(\sum_i \omega_i \mathcal{A}_{ij}\right)}_{\tilde{\omega}_j^\ell} \hat{x}_{j,t}. \end{aligned} \tag{152}$$

Thus

$$\begin{aligned} \hat{v}_t^{agg} &= \sum_j \tilde{\omega}_j^\ell \left[s_j(\hat{\Delta}_t + \hat{q}_t + \hat{b}_{j,t}) + e_j(\hat{q}_t + \hat{b}_{j,t-1})\right] \\ &= \left(\sum_j \tilde{\omega}_j^\ell s_j\right)\hat{\Delta}_t + \left(\sum_j \tilde{\omega}_j^\ell (s_j + e_j)\right)\hat{q}_t \\ &\quad + \sum_j \tilde{\omega}_j^\ell s_j \hat{b}_{j,t} + \sum_j \tilde{\omega}_j^\ell e_j \hat{b}_{j,t-1}. \end{aligned} \tag{153}$$

Define

$$\bar{\mu} = \sum_j \tilde{\omega}_j^\ell s_j, \qquad \bar{\alpha} = \sum_j \tilde{\omega}_j^\ell (s_j + e_j), \qquad \Lambda_j^{(0)} = \tilde{\omega}_j^\ell s_j, \qquad \Lambda_j^{(1)} = \tilde{\omega}_j^\ell e_j. \tag{154}$$

Substituting into (150) gives

$$\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \bar{\mu}\hat{\Delta}_t + \bar{\alpha}\hat{q}_t + \mathbf{\Lambda}^{(0)}\hat{\mathbf{b}}_t^{FI} + \mathbf{\Lambda}^{(1)}\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}). \tag{155}$$

A.4.5 Spread and long-bond return closure

Let $\mathcal{R} \equiv \beta/\beta_B > 1$. The saver Euler equation gives

$$\mathbb{E}_t[M_{t,t+1}^N R_t^s] = 1. \tag{156}$$

The steady spread is $\bar{\Delta} = \mathcal{R} - 1$. A first-order approximation of (74), using (156), gives

$$\hat{\Delta}_t = \frac{\mathcal{R}}{\mathcal{R}-1}\left(\mathbb{E}_t\hat{r}_{t+1}^b - \hat{r}_t^s\right). \tag{157}$$

Log-linearizing (58) around $\bar{Q}$ gives

$$\hat{r}_{t+1}^b = \frac{\kappa\bar{Q}}{1+\kappa\bar{Q}}\hat{q}_{t+1} - \hat{q}_t. \tag{158}$$

Under the steady-state borrower pricing condition,

$$\frac{\kappa\bar{Q}}{1+\kappa\bar{Q}} = \beta_B\kappa,$$

so

$$\hat{r}_{t+1}^b \approx \beta_B\kappa\hat{q}_{t+1} - \hat{q}_t. \tag{159}$$

B Proofs for Section 2

B.1 Proof of Proposition 1 (Value Dissipation)

Proposition (Value Dissipation, Formal Statement)**.** *For any cross-holdings matrix* $\mathbf{C} \in \mathbb{R}_+^{n\times n}$ *satisfying (i)* $C_{ii} = 0$ *for all* i*, (ii)* $\hat{C}_{ii} \equiv 1 - \sum_j C_{ji} \in (0,1]$ *for all* i*, (iii)* $\rho(\mathbf{C}) < 1$*, and any* $\tau \in [0,1)$*, the dependency matrix* $\mathbf{A}(\tau) = \hat{\mathbf{C}}\,[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ *satisfies*

$$\sum_{i\in\mathcal{N}} A_{ij}(\tau) = 1 - \tau\sum_{i\in\mathcal{N}}\left[\mathbf{C}\,(\mathbf{I}-(1-\tau)\mathbf{C})^{-1}\right]_{ij} \le 1 \quad \textit{for all } j \in \mathcal{N},$$

with equality if and only if $\tau = 0$ *or* $\mathbf{C} = \mathbf{0}$*. Equivalently, writing* $\mathbf{M}(\tau) \equiv [\mathbf{I}-(1-\tau)\mathbf{C}]^{-1}$*:*

$$\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}' - \tau\,\mathbf{1}'\mathbf{C}\,\mathbf{M}(\tau).$$

Proof. Write $\mathbf{M} = \mathbf{M}(\tau)$ for brevity. We compute $\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}'\hat{\mathbf{C}}\,\mathbf{M}$.

The i-th element of the row vector $\mathbf{1}'\hat{\mathbf{C}}$ is $\hat{C}_{ii} = 1 - \sum_j C_{ji}$, so in vector form:

$$\mathbf{1}'\hat{\mathbf{C}} = \mathbf{1}' - \mathbf{1}'\mathbf{C} = \mathbf{1}'(\mathbf{I} - \mathbf{C}).$$

Now decompose $\mathbf{I} - \mathbf{C}$ using the identity $\mathbf{I} - \mathbf{C} = [\mathbf{I} - (1-\tau)\mathbf{C}] - \tau\mathbf{C} = \mathbf{M}^{-1} - \tau\mathbf{C}$:

$$\mathbf{1}'\hat{\mathbf{C}} = \mathbf{1}'(\mathbf{M}^{-1} - \tau\mathbf{C}) = \mathbf{1}'\mathbf{M}^{-1} - \tau\,\mathbf{1}'\mathbf{C}.$$

Multiplying on the right by $\mathbf{M}$:

$$\mathbf{1}'\mathbf{A}(\tau) = \mathbf{1}'\hat{\mathbf{C}}\,\mathbf{M} = \mathbf{1}' - \tau\,\mathbf{1}'\mathbf{C}\,\mathbf{M}.$$

Sign of the dissipation term. Since $\mathbf{C} \geq \mathbf{0}$ elementwise and $\mathbf{M} = \sum_{k=0}^{\infty}[(1-\tau)\mathbf{C}]^k \geq \mathbf{I}$ elementwise, the product $\mathbf{CM} \geq \mathbf{C} \geq \mathbf{0}$. For any j such that $\sum_i C_{ij} > 0$ (i.e., bank j is held by at least one other bank), we have $\sum_i [\mathbf{CM}]_{ij} > 0$, so $\tau \sum_i [\mathbf{CM}]_{ij} > 0$ when $\tau > 0$. Hence $\sum_i A_{ij}(\tau) < 1$ strictly.

Equality conditions. When $\tau = 0$, the dissipation term vanishes identically. When $\mathbf{C} = \mathbf{0}$, $\mathbf{M} = \mathbf{I}$ and $\mathbf{CM} = \mathbf{0}$, so the dissipation term again vanishes. In all other cases with $\tau > 0$ and $\mathbf{C} \neq \mathbf{0}$, the inequality is strict. □

Remark 1. *The dissipation result applies to the dependency matrix $\mathbf{A}(\tau) = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$, where $\hat{\mathbf{C}}$ is the outside ownership matrix defined in Section 2.1.1. As emphasized in Section 2.1.3, this result depends only on the general properties of $\mathbf{A}$ (non-negativity, column sums bounded by unity) and not on the specific cross-holdings structure used to derive it. Any micro-foundation that generates a dependency matrix with column sums strictly less than one when frictions are present will produce the same qualitative conclusion.*

B.2 Proof of Proposition 2 (Value Feedback Channel)

Proposition (Value Feedback Channel, Formal Statement)**.** *Under Assumption 1 and the stability condition in Section 2.1.2, the semi-elasticity of bank i's credit supply with respect to the aggregate credit spread Δ_t is:*

$$\mathcal{Z}_{i,t} \equiv \frac{\partial \ln B^*_{i,t}}{\partial \Delta_t} = \frac{\phi_{i,t}\,A_{ii}}{1 - \phi_{i,t}\,A_{ii}\,\Delta_t}.$$

This semi-elasticity is (i) strictly positive, (ii) strictly increasing in the dependency index A_{ii}, and (iii) strictly convex in both A_{ii} and Δ_t.

Proof. Taking the natural logarithm of the optimal credit supply (14):

$$\ln B^*_{i,t} = \ln\big[\phi_{i,t}(A_{ii}\,\bar{x}_{i,t} + \bar{v}_{i,t})\big] - \ln Q_t - \ln\big(1 - \phi_{i,t}\,A_{ii}\,\Delta_t\big).$$

Differentiating with respect to Δ_t (treating $\bar{x}_{i,t}$, $\bar{v}_{i,t}$, Q_t, and $\phi_{i,t}$ as exogenous in partial equilibrium):

$$\mathcal{Z}_{i,t} = \frac{\partial \ln B^*_{i,t}}{\partial \Delta_t} = \frac{\phi_{i,t}\, A_{ii}}{1 - \phi_{i,t}\, A_{ii}\, \Delta_t}.$$

Part (i): Positivity. The denominator is positive by the stability condition in Section 2.1.2 and the numerator is positive since $\phi_{i,t} > 0$ and $A_{ii} > 0$.

Part (ii): Monotonicity in A_{ii}. Differentiating with respect to A_{ii}:

$$\frac{\partial \mathcal{Z}_{i,t}}{\partial A_{ii}} = \frac{\phi_{i,t}}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^2} > 0.$$

Part (iii): Convexity. Differentiating the semi-elasticity a second time:

$$\frac{\partial^2 \mathcal{Z}_{i,t}}{\partial A_{ii}^2} = \frac{2\, \phi_{i,t}^2\, \Delta_t}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^3} > 0,$$

and similarly for Δ_t:

$$\frac{\partial^2 \mathcal{Z}_{i,t}}{\partial \Delta_t^2} = \frac{2\, \phi_{i,t}^2\, A_{ii}^2}{(1 - \phi_{i,t}\, A_{ii}\, \Delta_t)^3} > 0.$$

□

B.3 Proof of Proposition 3 (Policy-Induced Credit Reallocation)

Proposition (Policy-Induced Credit Reallocation, Formal Statement)**.** *Under Assumption 1 and the stability condition in Section 2.1.2, the sensitivity of market concentration to an aggregate credit spread shock is:*

$$dHHI_t = 2 \cdot \mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t}) \cdot d\Delta_t.$$

Since $\mathcal{Z}_{i,t}$ is strictly increasing in A_{ii} (Proposition 2), concentration rises in response to accommodative policy ($d\Delta_t > 0$) if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$.

Proof. The change in bank i's market share satisfies:

$$d \ln \theta_{i,t} = d \ln B_{i,t} - d \ln B_t = \left(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\right) d\Delta_t,$$

where $\bar{\mathcal{Z}}_t \equiv \mathbb{E}_{\theta,t}[\mathcal{Z}_{j,t}]$ is the share-weighted average semi-elasticity. The second equality uses the fact that the growth rate of aggregate credit is the share-weighted average of individual growth rates:

$$d \ln B_t = \sum_{j \in \mathcal{N}} \frac{B_{j,t}}{B_t}\, d \ln B_{j,t} = \sum_{j \in \mathcal{N}} \theta_{j,t}\, \mathcal{Z}_{j,t}\, d\Delta_t = \bar{\mathcal{Z}}_t\, d\Delta_t.$$

Differentiating $HHI_t = \sum_i \theta_{i,t}^2$:

$$\begin{aligned} dHHI_t &= 2\sum_i \theta_{i,t}\, d\theta_{i,t} = 2\sum_i \theta_{i,t}^2 \big(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\big)\, d\Delta_t \\ &= 2\Big[\mathbb{E}_{\theta,t}[\theta_{i,t}\, \mathcal{Z}_{i,t}] - \mathbb{E}_{\theta,t}[\theta_{i,t}]\, \mathbb{E}_{\theta,t}[\mathcal{Z}_{i,t}]\Big]\, d\Delta_t \\ &= 2\,\mathrm{Cov}_{\theta,t}(\theta_{i,t},\, \mathcal{Z}_{i,t})\, d\Delta_t. \end{aligned}$$

The final claim follows from the monotonicity of $\mathcal{Z}_{i,t}$ in A_{ii} (Proposition 2): a positive covariance between $\theta_{i,t}$ and $\mathcal{Z}_{i,t}$ obtains if and only if $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii}) > 0$, since $\mathcal{Z}_{i,t}$ is a strictly increasing transformation of A_{ii} (holding $\phi_{i,t}$ constant). □

Corollary (Market Share Reallocation, Formal Statement)**.** *In response to a marginal increase in the credit spread $d\Delta_t > 0$, bank i's market share grows ($d\theta_{i,t} > 0$) if and only if its semi-elasticity exceeds the share-weighted average:*

$$d\ln\theta_{i,t} = \big(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t\big)\, d\Delta_t.$$

Since $\mathcal{Z}_{i,t}$ is increasing in A_{ii}, capital flows toward banks with higher dependency indices.

Proof. By definition, $\ln\theta_{i,t} = \ln B_{i,t} - \ln B_t$. Differentiating with respect to the aggregate shock:

$$d\ln\theta_{i,t} = d\ln B_{i,t} - d\ln B_t.$$

Substituting $d\ln B_{i,t} = \mathcal{Z}_{i,t}\, d\Delta_t$ and $d\ln B_t = \bar{\mathcal{Z}}_t\, d\Delta_t$ yields the result. The condition for reallocation follows directly from the sign of $(\mathcal{Z}_{i,t} - \bar{\mathcal{Z}}_t)$, which by Proposition 2 is positive if and only if A_{ii} exceeds the share-weighted average dependency index (holding $\phi_{i,t}$ constant). □

B.4 Proof of Proposition 5 (Symmetric vs. Asymmetric Networks)

Proposition (Symmetric vs. Asymmetric Networks, Formal Statement)**.** *Consider the class of networks parameterized by a uniform cross-holding intensity $\delta \in [0, 1)$ and a topology matrix $\mathbf{G}$ with $G_{ii} = 0$ and $\sum_j G_{ij} = 1$ for all i, so that $\mathbf{C} = \delta\mathbf{G}$.*

(a) Symmetric networks. *If $\mathbf{G}$ is vertex-transitive, then $A_{ii} = A_{jj}$ for all i, j, the semi-elasticities $\mathcal{Z}_{i,t}$ are equalized across banks conditional on common $\phi_{i,t}$, and $dHHI_t = 0$.*

(b) Asymmetric networks. *If banks occupy non-equivalent network positions, dependency indices generically differ across banks. The direction of reallocation is governed by $\mathrm{Cov}_{\theta,t}(\theta_{i,t}, A_{ii})$.*

Proof. If $\mathbf{G}$ is vertex-transitive, for any banks i and j there is a permutation matrix $\mathbf{P}_{ij}$ such that $\mathbf{P}_{ij}\mathbf{G}\mathbf{P}'_{ij} = \mathbf{G}$ and $\mathbf{P}_{ij}e_i = e_j$. Hence the same permutation leaves $\mathbf{C} = \delta\mathbf{G}$, outside ownership, and $[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ invariant. The diagonal entries of $\mathbf{A} = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1}$ are therefore identical. With common $\phi_{i,t}$, the semi-elasticities are identical and reallocation is zero. If banks are not in equivalent network positions, the outside ownership shares or the diagonal entries of the network inverse generically differ, so A_{ii} differs across banks and Proposition 3 gives the reallocation result. □

C Proofs for Section 4

C.1 Proof of Proposition 6 (Alignment Covariance)

Proposition (Alignment Covariance, Formal Statement). *Let $\Lambda_i^{(0)} \equiv \tilde{\omega}_i^{\ell} s_i$ be the contemporaneous credit-distribution loading defined in Section 3.4. Define bank i's* capacity intensity*,*

$$\kappa_i \equiv \frac{\Lambda_i^{(0)}}{\omega_i} = \frac{\tilde{\omega}_i^{\ell} s_i}{\omega_i},$$

its share-weighted average $\bar{\kappa} \equiv \mathbb{E}_\omega[\kappa_i]$, its normalized form $g_i \equiv \kappa_i/\bar{\kappa}$ (with $\mathbb{E}_\omega[g_i] = 1$), and the capacity wedge *$\xi_i \equiv g_i - 1$. Let $Z_i^m \equiv \partial \ln B_{i,t}/\partial m$ be bank i's lending semi-elasticity to the monetary shock m. Holding aggregate bank credit fixed, the contribution of cross-sectional credit reallocation to aggregate financial capacity satisfies*

$$\frac{d\Theta_t^{realloc}}{dm} = \mathrm{Cov}_\omega\big(\kappa_i,\, Z_i^m\big) = \bar{\kappa}\,\mathrm{Cov}_\omega\big(\xi_i,\, Z_i^m\big),$$

where the covariances are share-weighted across banks.

Proof. **Cross-sectional objects.** Recall from Section 3.4 that the contemporaneous credit-distribution loading is $\Lambda_i^{(0)} = \tilde{\omega}_i^{\ell} s_i$. The capacity intensity, its share-weighted average, the normalized form, and the capacity wedge are defined in the proposition statement above. The wedge is positive when bank i's contribution to aggregate financial capacity per unit of credit exceeds the cross-sectional average, and negative when bank i is below average. Note that $\bar{\kappa} = \sum_i \omega_i \kappa_i = \sum_i \Lambda_i^{(0)} > 0$ under the regularity conditions of Section 2.1.2.

Bank-level lending response. For the spread experiment of Proposition 2, $Z_i^m = \mathcal{Z}_{i,t} \cdot \partial\Delta_t/\partial m$. For richer experiments, Z_i^m aggregates the induced movements in spreads, bond prices, and balance-sheet positions through the credit-distribution law (36). Aggregate lending growth is the share-weighted average,

$$d\ln B_t = \mathbb{E}_\omega[Z_i^m]\,dm \equiv \bar{Z}^m\,dm,$$

and bank i's market share evolves according to

$$d \ln \theta_{i,t} = \left(Z_i^m - \bar{Z}^m\right) dm,$$

which is the log-share counterpart of the corollary to Proposition 3.

Aggregate capacity response. Use the identity $\hat{b}_{i,t} = \hat{b}_t^{FI} + \hat{\theta}_{i,t}$ in the credit-distribution loading $\Theta_t^B \equiv \sum_i \Lambda_i^{(0)} \hat{b}_{i,t}$:

$$\Theta_t^B = \Big(\sum_i \Lambda_i^{(0)}\Big) \hat{b}_t^{FI} + \sum_i \Lambda_i^{(0)} \hat{\theta}_{i,t} = \bar{\kappa}\, \hat{b}_t^{FI} + \sum_i \omega_i \, \kappa_i \, \hat{\theta}_{i,t}.$$

Differentiating with respect to the monetary shock m and substituting the share-evolution result,

$$\frac{d\Theta_t^B}{dm} = \bar{\kappa}\, \bar{Z}^m + \sum_i \omega_i \, \kappa_i \left(Z_i^m - \bar{Z}^m\right) = \bar{\kappa}\, \bar{Z}^m + \mathrm{Cov}_\omega(\kappa_i,\, Z_i^m).$$

The first term is the contribution from the *aggregate-credit* margin: total bank lending expands and capacity scales with $\bar{\kappa}$. The second term is the *cross-sectional reallocation* margin and equals the share-weighted covariance of the capacity intensity and the lending response.

Reallocation contribution. Holding aggregate bank credit fixed,

$$\frac{d\Theta_t^{\text{realloc}}}{dm} \equiv \sum_i \Lambda_i^{(0)} \, d\hat{\theta}_{i,t} = \mathrm{Cov}_\omega(\kappa_i,\, Z_i^m).$$

Substituting $\kappa_i = \bar{\kappa}(1 + \xi_i)$ and using $\mathrm{Cov}_\omega(\text{constant}, \cdot) = 0$,

$$\mathrm{Cov}_\omega(\kappa_i,\, Z_i^m) = \bar{\kappa}\, \mathrm{Cov}_\omega(\xi_i,\, Z_i^m),$$

which establishes the identity in the proposition statement and proves Proposition 6. □

Remark 2 (Special cases)**.** *The covariance is zero in three economically distinct cases. (i) If $Z_i^m = \bar{Z}^m$ for all i, monetary policy moves no credit across banks. (ii) If $\xi_i = 0$ for all i, every bank's contribution to aggregate capacity is proportional to its market share, and credit reallocation is aggregation-neutral. (iii) The degenerate-network limit $\mathbf{C} = \mathbf{0}$ implies $\tilde{\omega}_i^\ell = \omega_i$ and hence $\xi_i = 0$ for all i, recovering case (ii) and consistent with Proposition 10.*

C.2 Proof of Proposition 7 (Financial Reallocation and Output–Inflation Composition)

Proposition (Financial Reallocation and Output–Inflation Composition, Formal Statement)**.** *Consider a monetary shock and define*

$$\gamma_t \equiv \left.\frac{\partial \Theta_t^{net}}{\partial \hat{y}_t}\right|_{monetary\ shock}.$$

The Phillips Curve can be written, for the component of inflation that co-moves with output under this policy experiment, as

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \tilde{\kappa}_t \hat{y}_t + \tilde{u}_t, \qquad \tilde{\kappa}_t = \kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t.$$

If $\gamma_t > 0$, the co-movement of output with network financial capacity creates a marginal-cost offset to the inflationary effect of higher output. Comparing a network economy and a no-network economy under the same policy experiment, monetary transmission shifts further toward real activity and away from inflation in the network economy if and only if

$$\gamma_t^{\text{net}} > \gamma_t^{\text{no net}}.$$

This condition is more likely when banks with large policy responses also carry large effective network weights.

Proof. Start from the augmented Phillips Curve in equation (33):

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa_y \hat{y}_t - \Lambda_C \big(\bar{b}^{FI} \Theta_t^{\text{net}} + \bar{b}^{CB} \widehat{qe}_t\big) - \psi_a \hat{a}_t.$$

For the monetary experiment under consideration, decompose network financial capacity into the component that co-moves with output and the component orthogonal to output:

$$\Theta_t^{\text{net}} = \gamma_t \hat{y}_t + \Theta_t^{\perp}, \qquad \left.\frac{\partial \Theta_t^{\text{net}}}{\partial \hat{y}_t}\right|_{\text{monetary shock}} = \gamma_t.$$

Substituting this decomposition into the Phillips Curve and collecting terms proportional to $\hat{y}_t$ gives

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \big(\kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t\big) \hat{y}_t + \tilde{u}_t,$$

where

$$\tilde{u}_t = -\Lambda_C \bar{b}^{FI} \Theta_t^{\perp} - \Lambda_C \bar{b}^{CB} \widehat{qe}_t - \psi_a \hat{a}_t$$

collects terms not proportional to $\hat{y}_t$. This proves the stated representation with $\tilde{\kappa}_t = \kappa_y - \Lambda_C \bar{b}^{FI} \gamma_t$.

Because $\Lambda_C > 0$ and $\bar{b}^{FI} > 0$, a positive γ_t lowers the inflation response associated

with a given output movement relative to the case in which financial capacity is held fixed. Equivalently, the financial-capacity movement produces the supply-side credit offset described in the main text.

Now compare the network and no-network economies. Holding fixed the mechanical preference and pricing parameters $(\kappa_y, \Lambda_C, \bar{b}^{FI})$, the difference in the effective output coefficient is

$$\tilde{\kappa}_t^{\text{net}} - \tilde{\kappa}_t^{\text{no net}} = -\Lambda_C \bar{b}^{FI} \big(\gamma_t^{\text{net}} - \gamma_t^{\text{no net}}\big).$$

Hence

$$\tilde{\kappa}_t^{\text{net}} < \tilde{\kappa}_t^{\text{no net}} \quad \Longleftrightarrow \quad \gamma_t^{\text{net}} > \gamma_t^{\text{no net}}.$$

Since a smaller output coefficient means that a given output response is accompanied by a smaller inflation response, the condition is exactly the statement that monetary transmission shifts further toward real activity and away from inflation.

It remains to connect the condition to reallocation. From equation (52) and Proposition 6, the cross-sectional component of γ_t^{net} is proportional to the alignment covariance $\text{Cov}_\omega(\xi_i, Z_i^m)$. This covariance is positive precisely when high-response banks (high Z_i^m) are also banks with positive capacity wedges (high ξ_i, equivalently high $\tilde{\omega}_i^\ell$ relative to ω_i), which proves the alignment statement. □

C.3 Proof of Proposition 8 (Conditional Asymmetric Amplification)

Proposition (Conditional Asymmetric Network Amplification, Formal Statement)**.** *Let the output response of each policy instrument be decomposed into a direct household Euler-equation component and a bank balance-sheet component. Suppose the network multiplies the balance-sheet component of both instruments by a common factor $\rho > 1$, and suppose the interest-rate instrument has a positive direct household component that is invariant to the network. If QE loads only on the balance-sheet component in the comparison, then the output amplification ratio for QE exceeds the output amplification ratio for conventional interest-rate policy.*

Proof. Let $B_r > 0$ denote the no-network balance-sheet component of the output response to interest-rate policy, and let $B_q > 0$ denote the no-network balance-sheet component of the output response to QE. Let $H > 0$ denote the direct household Euler-equation component of interest-rate policy. This component is a household intertemporal-substitution response and is therefore independent of the financial network. Under the maintained assumption, the network multiplies balance-sheet components by the common factor $\rho > 1$:

$$B_r^{\text{net}} = \rho B_r, \qquad B_q^{\text{net}} = \rho B_q.$$

For QE, the output response in this comparison is entirely balance-sheet based, so its amplification ratio is

$$\mathcal{R}_q = \frac{B_q^{\text{net}}}{B_q} = \rho.$$

For interest-rate policy, the output response contains both the direct household bypass and the balance-sheet component. Its amplification ratio is

$$\mathcal{R}_r = \frac{H + \rho B_r}{H + B_r}.$$

Since $H > 0$ and $\rho > 1$,

$$H + \rho B_r < \rho H + \rho B_r = \rho(H + B_r),$$

so

$$\mathcal{R}_r = \frac{H + \rho B_r}{H + B_r} < \rho = \mathcal{R}_q.$$

The inequality is strict because the household bypass is positive. Thus the instrument with the larger balance-sheet share, QE, receives the larger overall network amplification even when the network multiplies the common balance-sheet component by the same factor for both instruments. □

C.4 Proof of Proposition 9 (Financial Memory Through Inherited Bond Holdings)

Proposition (Financial Memory Through Inherited Bond Holdings, Formal Statement)**.** *In the first-order system with fixed valuation-elasticity matrix $\mathcal{A}$ and fixed effective network weights $\tilde{\omega}^{\ell}$, inherited bank bond holdings affect current network financial capacity through the term $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ in equation (52). The cross-section of bank bond holdings evolves according to*

$$\hat{\mathbf{b}}_t^{FI} = \mathbf{P}\hat{\mathbf{b}}_{t-1}^{FI} + \textit{current shocks and prices}, \qquad \mathbf{P} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}).$$

Consequently, past monetary policy affects the future financial position from which new shocks are transmitted. The persistence of this inherited financial wedge is increasing in $\rho(\mathbf{P})$.

Proof. From equation (52), current network financial capacity can be written as

$$\Theta_t^{\text{net}} = \hat{\phi}_t^{agg} + \mathbf{\Gamma}'\hat{\boldsymbol{\phi}}_t + \mu^*\hat{\Delta}_t + \alpha^*\hat{q}_t + \mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI} - \eta_B(\hat{b}_t^{FI} - \hat{b}_{t-1}^{FI}),$$

where

$$\mathbf{\Psi}' = \mathbf{\Lambda}^{(1)} + \mathbf{\Gamma}'(\mathcal{A}\mathbf{E} + \eta_B\mathbf{I}).$$

The term $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$ is the direct effect of inherited cross-sectional bond holdings on current financial capacity. It is cross-sectional: for a fixed aggregate $\hat{b}_{t-1}^{FI}$, different allocations of inherited bond holdings across banks can generate different values of this term whenever the components of $\mathbf{\Psi}$ differ across banks.

The law of motion for credit in equation (36) can be written as

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathcal{A}\mathbf{S}]^{-1}\big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q\hat{q}_t + \boldsymbol{\alpha}_\Delta\hat{\Delta}_t\big] + \mathbf{P}\hat{\mathbf{b}}_{t-1}^{FI},$$

with $\mathbf{P}$ as stated. Iterating forward from an inherited distribution at date $t-1$ gives

$$\hat{\mathbf{b}}_{t+h}^{FI} = \mathbf{P}^{h+1}\hat{\mathbf{b}}_{t-1}^{FI} + \sum_{m=0}^{h}\mathbf{P}^m\big(\text{current shocks and prices at } t+h-m\big).$$

Thus the effect of an inherited allocation on future credit distributions is governed by powers of $\mathbf{P}$.

By the Gelfand formula,

$$\lim_{h\to\infty}\|\mathbf{P}^h\|^{1/h} = \rho(\mathbf{P})$$

for any matrix norm. Hence a larger spectral radius implies slower decay of inherited cross-sectional positions, conditional on stability $\rho(\mathbf{P}) < 1$. Since inherited positions enter financial capacity through $\mathbf{\Psi}'\hat{\mathbf{b}}_{t-1}^{FI}$, this establishes that the persistence of the inherited financial wedge is increasing in $\rho(\mathbf{P})$.

Finally, because $\mathcal{A}$ and $\tilde{\omega}^\ell$ are fixed around the steady state in the first-order approximation, the result is not a change in local structural coefficients. It is a shift in the inherited financial wedge and therefore in the financial position from which future policy shocks are transmitted. □

C.5 Proof of Proposition 10 (Degenerate Network Limit)

Proposition (Degenerate Network Limit, Formal Statement)**.** *When* $\mathbf{C} = \mathbf{0}$*, the level dependency matrix is* $\mathbf{A} = \mathbf{I}$ *and the valuation-elasticity matrix is* $\mathcal{A} = \mathbf{I}$*. The effective network weights collapse to market shares,*

$$\tilde{\omega}_j^\ell = \omega_j,$$

and the credit multiplier reduces to the diagonal matrix

$$[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}.$$

If banks are ex ante identical, the cross-sectional credit distribution is irrelevant for aggregation and the model collapses to a representative-intermediary New Keynesian system

augmented by a scalar financial-capacity state.

Proof. Dependency and valuation matrices. If $\mathbf{C} = \mathbf{0}$, then outside ownership is complete: $\hat{\mathbf{C}} = \mathbf{I}$. Therefore

$$\mathbf{A} = \hat{\mathbf{C}}[\mathbf{I} - (1-\tau)\mathbf{C}]^{-1} = \mathbf{I}.$$

Since $\mathbf{v}_t = \mathbf{x}_t$ in this limit, $\bar{v}_i = \bar{x}_i$ for all i. The valuation-elasticity matrix satisfies

$$\mathcal{A}_{ij} = \frac{A_{ij}\bar{x}_j}{\bar{v}_i} = \frac{\delta_{ij}\bar{x}_j}{\bar{x}_i} = \delta_{ij},$$

so $\mathcal{A} = \mathbf{I}$.

Effective weights. Using the definition of the effective network weight,

$$\tilde{\omega}_j^{\ell} = \sum_i \omega_i \mathcal{A}_{ij} = \sum_i \omega_i \delta_{ij} = \omega_j.$$

Thus there is no distinction between a bank's market share and its aggregate consequence weight.

Credit multiplier. Substituting $\mathcal{A} = \mathbf{I}$ into the credit distribution block gives

$$\hat{\mathbf{b}}_t^{FI} = [(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}\Big[\hat{\boldsymbol{\phi}}_t + \boldsymbol{\alpha}_q \hat{q}_t + \boldsymbol{\alpha}_\Delta \hat{\Delta}_t + (\mathbf{E} + \eta_B \mathbf{I})\hat{\mathbf{b}}_{t-1}^{FI}\Big].$$

Because $\mathbf{S}$ and $\mathbf{E}$ are diagonal, the multiplier $[(1+\eta_B)\mathbf{I} - \mathbf{S}]^{-1}$ is diagonal. Bank-level lending dynamics are therefore uncoupled: a shock to bank j affects bank j's credit through its own balance sheet, but there is no network propagation to other banks.

Aggregation. With $\tilde{\omega}_j^{\ell} = \omega_j$, the aggregation coefficients in equation (41) become

$$\bar{\mu} = \sum_j \omega_j s_j, \qquad \bar{\alpha} = \sum_j \omega_j (s_j + e_j), \qquad \Lambda_j^{(0)} = \omega_j s_j, \qquad \Lambda_j^{(1)} = \omega_j e_j.$$

If banks are ex ante identical, then $s_j = s$, $e_j = e$, and $\phi_{j,t} = \phi_t$ for all j. Since $\sum_j \omega_j = 1$,

$$\bar{\mu} = s, \qquad \bar{\alpha} = s + e, \qquad \boldsymbol{\Lambda}^{(0)}\hat{\mathbf{b}}_t^{FI} = s\sum_j \omega_j \hat{b}_{j,t} = s\hat{b}_t^{FI},$$

and similarly

$$\boldsymbol{\Lambda}^{(1)}\hat{\mathbf{b}}_{t-1}^{FI} = e\hat{b}_{t-1}^{FI}.$$

Thus the network aggregation equation depends on the banking sector only through the scalar aggregate $\hat{b}_t^{FI}$ and the scalar financial capacity $\Theta_t^{\text{net}} = \hat{q}_t + \hat{b}_t^{FI}$.

The cross-sectional distribution is therefore irrelevant in the degenerate, ex ante identical limit. Substituting the scalar financial capacity state into the IS curve and the Phillips Curve yields the representative-intermediary New Keynesian system augmented by one financial-capacity state, as claimed. □